

# PROJECT REPORT

On

## RECOMMENDATION SYSTEMS FOR PERSONALIZED CUSTOMER EXPERIENCE IN THE E-COMMERCE INDUSTRY

In partial fulfillment of the requirements for the award of the degree of

**Master of Business Administration (MBA)**

in Data Science and Analysis

**Submitted By**

**Under the Supervision of**

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**JULY 2026**

# DECLARATION

I, Vikas Jadaun, USN: 2420500108, hereby declare that the project report entitled **"Recommendation Systems for Personalized Customer Experience in the E-Commerce Industry"** submitted to the Centre for Distance and Online Education, Sharda University, Greater Noida, in partial fulfillment of the requirements for the award of the degree of Master of Business Administration in Data Science and Analysis, is a record of original and independent research work carried out by me under the supervision and guidance of Dr. Aashima.

I further declare that this report or any part thereof has not been submitted previously to any other university or institution for the award of any degree, diploma, fellowship, or similar title. All sources of information and assistance received in the course of this research have been duly acknowledged and cited in accordance with the APA 7th Edition referencing style.

I also declare that the content of this report is original, and the plagiarism similarity index is below 10% and AI similarity is below 20%, in compliance with the academic integrity guidelines of Sharda University.

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# CERTIFICATE

This is to certify that the project report entitled "**Recommendation Systems for Personalized Customer Experience in the E-Commerce Industry**" submitted by **Vikas Jadaun** (USN: 2420500108), in partial fulfillment of the requirements for the award of the degree of Master of Business Administration in Data Science and Analysis, Centre for Distance and Online Education, Sharda University, Greater Noida, is a record of bonafide research work carried out by the candidate under my supervision and guidance.

The report has been prepared by the candidate himself/herself and the content of this report is original. To the best of my knowledge and belief, this work has not been submitted, either in part or in full, to any other university or institution for the award of any degree, diploma, or certificate.

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# ABSTRACT

The rapid growth of e-commerce has fundamentally transformed the way consumers discover, evaluate, and purchase products online. With catalogues expanding into millions of items across diverse categories, customers increasingly expect shopping experiences that feel personal and relevant. Personalization has emerged as a critical differentiator, and recommendation systems sit at the heart of this transformation. This study investigates how different recommendation system architectures perform in the e-commerce context, with the goal of determining whether Collaborative Filtering, Content-Based Filtering, or a weighted Hybrid approach delivers the best balance of prediction accuracy, recommendation relevance, and catalogue coverage.

The research employs a quantitative experimental design using a synthetic e-commerce dataset comprising 1,000 customers, 455 products across 8 categories, and 49,058 interaction records (views, cart additions, and purchases) with 22,174 explicit ratings. The dataset exhibits a sparsity of 94.9%, representative of real-world e-commerce interaction matrices. Four recommendation models were developed and evaluated: a non-personalized Popularity Baseline, a Collaborative Filtering model using Truncated Singular Value Decomposition (SVD) with mean-centering, a Content-Based Filtering model using cosine similarity over product feature vectors, and a weighted Hybrid model blending both approaches.

Models were evaluated on a held-out test set using Root Mean Squared Error (RMSE) for prediction accuracy, Precision@10, Recall@10, and F1@10 for ranking quality, and Catalogue Coverage for diversity. The Collaborative Filtering model achieved the lowest RMSE of 1.5748, significantly outperforming both the Content-Based model (1.7107) and the Popularity Baseline (1.6557). The Hybrid model matched the CF model's RMSE at 1.5748 while demonstrating superior catalogue coverage (85.3%) compared to Content-Based Filtering (46.8%). All personalized models dramatically outperformed the Popularity Baseline on coverage (89.0% and 85.3% vs. 2.2%). The study also found that Frequent and VIP customer segments, representing only 35.7% of the customer base, accounted for 80.5% of total purchase volume (Chi-square = 30006.04,  $p < 0.001$ ), confirming highly concentrated purchasing behaviour.

The findings provide evidence-based guidance for e-commerce businesses on selecting recommendation architectures, considering trade-offs between accuracy, relevance, and diversity. The study confirms that Collaborative Filtering excels when sufficient user history exists, Content-Based Filtering provides utility for cold-start scenarios, and Hybrid approaches offer the most balanced overall performance. These insights are particularly valuable for e-commerce teams deciding between building in-house systems or adopting third-party platforms, and for organizations grappling with the cold-start challenge of

serving new customers and promoting new products.

# INTRODUCTION

## 1.1 Background of the Study

The e-commerce industry has undergone a paradigm shift over the past decade, evolving from simple online storefronts into sophisticated digital marketplaces that serve millions of consumers simultaneously. Global e-commerce sales reached approximately \$5.8 trillion in 2023 and are projected to surpass \$8 trillion by 2027, driven by increasing internet penetration, smartphone adoption, and changing consumer preferences. In India specifically, the e-commerce market has experienced explosive growth, expanding from approximately \$30 billion in 2019 to over \$70 billion in 2023, with projections indicating a trajectory toward \$160 billion by 2028. This growth has been fuelled by factors including the Digital India initiative, the rise of UPI-based digital payments, improved logistics infrastructure, and a young, tech-savvy demographic that is increasingly comfortable transacting online.

As the e-commerce landscape has matured, a critical challenge has emerged: the problem of information overload. Leading platforms such as Amazon, Flipkart, and Myntra now host catalogues containing tens of millions of products across thousands of categories. A customer searching for a smartphone, for instance, may be confronted with hundreds of options varying in brand, price, features, and specifications. The cognitive burden of manually evaluating these options is substantial, and research in consumer behaviour has consistently shown that when faced with excessive choice, customers are more likely to experience decision paralysis, abandon their shopping journey, or default to the most visible or popular option rather than the one that best matches their needs.

Recommendation systems have emerged as the primary technological solution to this challenge. These systems leverage data on customer behaviour, product attributes, and historical interactions to predict which products a given customer is most likely to be interested in, and present these personalized suggestions at various touchpoints in the shopping journey. The impact of effective recommendation systems on business outcomes is well-documented: Amazon has reported that its recommendation engine contributes to approximately 35% of total revenue, while Netflix estimates that its system saves the company over \$1 billion annually by reducing customer churn through improved content discovery.

The evolution of recommendation systems has followed several distinct paradigms. The earliest approaches were non-personalized, relying on aggregate popularity metrics to recommend the same set of trending products to all customers. While simple to implement, these systems fail to account for individual preferences and provide limited value beyond what a bestseller list might offer. The next major advancement was

Collaborative Filtering, which harnesses the collective wisdom of the user base by identifying customers with similar interaction patterns and recommending products that those similar customers have liked. This approach, pioneered by Goldberg et al. (1992) and later refined by Sarwar et al. (2001) through matrix factorization techniques, remains one of the most widely deployed recommendation methods in industry.

Content-Based Filtering offers a complementary approach, focusing on the attributes of products themselves rather than on user-to-user similarities. By constructing a profile of each customer's preferences based on the features of products they have previously interacted with, content-based systems can recommend items that share similar characteristics. This approach is particularly valuable for addressing the cold-start problem for new users, where collaborative methods struggle due to the absence of interaction history. However, content-based systems tend to suffer from over-specialization, recommending items that are too similar to what the user already knows, limiting serendipitous discovery.

Hybrid recommendation systems, which combine multiple recommendation strategies, have been proposed as a way to leverage the strengths of each approach while mitigating their individual weaknesses. Burke (2002) provided one of the earliest taxonomies of hybrid systems, identifying several architectural patterns including weighted, switching, mixed, feature combination, and cascade designs. The weighted hybrid approach, which linearly combines the scores from multiple models, is among the most commonly implemented due to its simplicity and effectiveness.

Despite the extensive body of research on recommendation systems, several gaps remain in the literature, particularly in the context of e-commerce. Many existing studies evaluate recommendation approaches in isolation, making it difficult to draw fair comparisons across methods under controlled conditions. Furthermore, much of the academic research has focused on the MovieLens dataset or similar movie/music domains, which differ significantly from e-commerce in terms of product characteristics, interaction patterns, and the importance of price and category attributes. There is also a relative scarcity of research that examines how different recommendation models perform across distinct customer segments, an aspect that is crucial for developing targeted personalization strategies in practice.

This study addresses these gaps by conducting a controlled, side-by-side comparison of three key recommendation approaches, a Collaborative Filtering model using Truncated Singular Value Decomposition, a Content-Based Filtering model using cosine similarity over product features, and a weighted Hybrid model combining both, against a non-personalized Popularity Baseline. The evaluation is conducted using a synthetic e-commerce dataset designed to reflect the messy realities of real-world data, including long-tail product popularity, heterogeneous customer segments, and significant sparsity in the interaction matrix. By evaluating all models on an identical test set using a

comprehensive suite of metrics, this research provides a rigorous and practically relevant comparison that can inform both academic understanding and business decision-making.

### **1.1.1 The Indian E-Commerce Landscape**

The Indian e-commerce market deserves special attention as one of the fastest-growing digital commerce markets globally. With over 800 million internet users and 750 million smartphone users as of 2024, India represents a massive opportunity for e-commerce platforms. The market is characterized by several unique features that influence the design and deployment of recommendation systems.

First, the Indian market is highly diverse, with significant variation in language, culture, income levels, and consumption patterns across regions. A recommendation system that works well for urban consumers in Bangalore may not be effective for rural consumers in Uttar Pradesh. This diversity necessitates recommendation approaches that can account for geographic and demographic variation, as explored in this study through the inclusion of location as a customer attribute and the analysis of location-category interaction patterns.

Second, the Indian e-commerce market is dominated by a few large platforms, including Flipkart (owned by Walmart), Amazon India, and emerging players like Reliance JioMart and Tata Neu. These platforms compete intensely on personalization, with recommendation quality being a key differentiator. The competitive landscape drives continuous innovation in recommendation technology, making the study of recommendation systems particularly relevant in the Indian context.

Third, the Indian market has unique seasonal patterns driven by festivals such as Diwali, Eid, and regional new year celebrations, as well as events like the Big Billion Days and Great Indian Festival sales. These seasonal spikes create temporal patterns in customer behaviour that recommendation systems can leverage. While this study does not explicitly model seasonal effects, the temporal distribution of interactions in the dataset reflects the general growth and fluctuation patterns observed in the Indian e-commerce market.

Fourth, the adoption of digital payments in India has been transformed by the Unified Payments Interface (UPI), which has made online transactions fast, convenient, and accessible to a broad population. The ease of digital payments has reduced friction in the purchase process, leading to higher conversion rates and more interaction data for recommendation systems to leverage. The growth of UPI transactions from 1 billion per month in 2020 to over 10 billion per month in 2024 illustrates the rapid digitization of the Indian payment ecosystem.

### **1.1.2 The Role of Data Science in E-Commerce Personalization**

Data science has become the backbone of e-commerce personalization, providing the methodological tools and analytical frameworks that enable recommendation systems to

operate at scale. The intersection of data science and e-commerce encompasses several key areas, each contributing to the overall goal of delivering personalized customer experiences.

**Predictive Analytics:** At its core, a recommendation system is a predictive analytics application, forecasting which products a customer is likely to be interested in based on historical data. The predictive models used in recommendation, including the matrix factorization and similarity-based approaches examined in this study, are fundamental data science techniques adapted to the specific challenges of the recommendation problem.

**Customer Segmentation:** Data science provides the clustering and segmentation techniques that enable e-commerce platforms to group customers based on behaviour, preferences, and demographics. The customer segmentation used in this study (New, Occasional, Frequent, VIP) is a simple example of this approach, but more sophisticated segmentation methods, including k-means clustering, hierarchical clustering, and model-based clustering, are widely used in industry.

**Big Data Processing:** E-commerce platforms generate massive volumes of data daily, including click streams, transaction records, search queries, and customer reviews. Data science provides the tools and frameworks for processing, storing, and analyzing these large datasets, enabling recommendation systems to operate on up-to-date data at scale.

**A/B Testing and Experimentation:** Data science methodologies, particularly experimental design and statistical inference, are essential for evaluating the impact of recommendation systems in production. A/B testing allows e-commerce platforms to compare different recommendation approaches on live traffic, measuring their impact on business metrics such as click-through rate, conversion rate, and revenue.

**Machine Learning Operations (MLOps):** The deployment and maintenance of recommendation models in production requires MLOps practices, including model versioning, monitoring, retraining pipelines, and rollback mechanisms. Data science provides the framework for managing the full lifecycle of recommendation models, from development to deployment to monitoring to retirement.

## 1.2 Research Problem Statement

The core problem addressed by this study can be stated as follows: In the e-commerce context, where product catalogues are large, customer interaction data is sparse, and user preferences are heterogeneous, which recommendation system architecture, Collaborative Filtering, Content-Based Filtering, or a Hybrid combination of both, delivers the best balance of prediction accuracy, recommendation relevance, and catalogue coverage? Furthermore, how do these personalized approaches compare to a non-personalized popularity-based baseline, and how does their performance vary across

different customer segments?

This problem is significant because e-commerce businesses face a real and consequential decision when selecting or building a recommendation system. The choice of architecture influences not only the accuracy of individual predictions but also the diversity of products surfaced to customers, the system's ability to handle new users and products (the cold-start problem), and the computational resources required for deployment. A wrong choice can lead to suboptimal customer experiences, reduced engagement, and lost revenue. Conversely, an informed decision based on empirical evidence can enhance customer satisfaction, increase conversion rates, and improve overall business performance.

The problem is further complicated by the inherent trade-offs between different evaluation criteria. A model that achieves the lowest prediction error (RMSE) may not necessarily produce the most relevant top-N recommendations (Precision@10, Recall@10), and a model that recommends highly relevant items may do so from a narrow subset of the catalogue, limiting exposure to the long-tail products that are crucial for niche customer satisfaction and inventory turnover. Understanding these trade-offs and how they manifest across different architectures and customer segments is essential for making well-rounded recommendations.

### 1.3 Research Objectives

This study pursues the following five objectives, each designed to address a specific aspect of the research problem:

**Objective 1:** To analyse customer interactions and transactional data, breaking down purchase patterns and engagement across at least four customer segments: New, Occasional, Frequent, and VIP, in order to understand the distribution of purchasing behaviour and identify concentration patterns within the e-commerce dataset.

**Objective 2:** To build a Collaborative Filtering recommendation model using matrix factorization via Truncated Singular Value Decomposition (SVD), and to demonstrate that it reduces rating prediction error (RMSE) compared to a non-personalized popularity-based baseline approach.

**Objective 3:** To construct a Content-Based Filtering recommendation model centred on product attributes including category, sub-category, price, brand, and average rating, and to evaluate its performance using Precision@10 and Recall@10 on a held-out test dataset.

**Objective 4:** To develop a Hybrid recommendation model by blending the Collaborative Filtering and Content-Based Filtering approaches through a weighted combination, and to compare its performance, measured by RMSE, Precision@10, Recall@10, and catalogue

coverage, against the individual models and the popularity baseline.

**Objective 5:** To provide practical, evidence-based recommendations for e-commerce businesses regarding the selection and deployment of recommendation systems, grounded strictly in the quantitative findings of the study and considering the trade-offs between accuracy, relevance, and diversity.

## 1.4 Research Questions

The study is guided by four research questions, each corresponding to a testable hypothesis:

**RQ1:** What behavioural and transactional patterns characterize customer engagement and purchase concentration across customer segments in the e-commerce dataset?

**RQ2:** How does a Collaborative Filtering model perform in predicting customer preferences relative to a Content-Based Filtering model, as measured by Root Mean Squared Error (RMSE)?

**RQ3:** Does a Hybrid recommendation approach outperform standalone Collaborative and Content-Based models in top-N recommendation relevance (Precision@10, Recall@10) and catalogue coverage?

**RQ4:** Do personalized recommendation approaches (Collaborative Filtering, Content-Based Filtering, and Hybrid) provide a measurable improvement over non-personalized, popularity-based recommendations across evaluation metrics?

## 1.5 Research Hypotheses

The following hypotheses, derived from the research questions, form the empirical backbone of this study. Each hypothesis is stated in a testable form and will be evaluated using quantitative model-evaluation results presented in Chapter 4.

| Hypothesis | Statement                                                                                                                                                                       |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| H1         | Frequent and VIP customer segments account for a significantly disproportionate share of total purchase volume relative to their share of the total customer base.              |
| H2         | The Collaborative Filtering model achieves a significantly lower Root Mean Squared Error (RMSE) in rating prediction than the Content-Based Filtering model.                    |
| H3         | The Hybrid model achieves significantly higher Precision@10, Recall@10, and catalogue coverage than either standalone Collaborative Filtering or Content-Based Filtering model. |



| Hypothesis | Statement                                                                                                                                                                                         |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| H4         | All three personalized models (Collaborative Filtering, Content-Based Filtering, and Hybrid) achieve significantly higher Precision@10 and Recall@10 than a non-personalized popularity baseline. |

Hypotheses H1 through H4 are tested using the quantitative model-evaluation results presented in Chapter 4, drawing on statistical comparison of segment-level purchase shares (H1) and model performance metrics including RMSE, Precision@10, Recall@10, and catalogue coverage (H2 through H4), computed on a held-out test dataset.

## 1.6 Significance of the Study

The significance of this research can be understood from both academic and practical perspectives. From an academic standpoint, this study contributes to the existing literature on recommendation systems by providing a controlled, side-by-side comparison of three major recommendation paradigms under identical experimental conditions. Much of the existing research evaluates recommendation approaches in isolation, making cross-method comparison difficult due to differences in datasets, evaluation protocols, and preprocessing steps. By evaluating all models on the same dataset, with the same train-test split, and using a consistent suite of metrics, this study offers a fair and directly comparable assessment of relative performance.

Furthermore, the study extends the literature by examining how recommendation performance varies across customer segments. Most existing studies treat the user population as homogeneous, reporting aggregate metrics that may obscure important segment-level differences. By segmenting customers into New, Occasional, Frequent, and VIP categories and evaluating model performance within each segment, this research provides a more nuanced understanding of which recommendation approaches work best for which types of customers, an insight that is directly relevant to the design of targeted personalization strategies.

From a practical perspective, the findings of this study offer e-commerce businesses evidence-based guidance on selecting recommendation system architectures. The trade-offs between accuracy, relevance, and coverage that are quantified in this study can inform decisions about whether to build a recommendation system in-house, adopt a third-party platform, or deploy a hybrid approach. The segment-level analysis is particularly valuable for businesses that serve diverse customer populations and need to tailor their personalization strategies accordingly. For example, the findings can help a business determine whether to rely on Collaborative Filtering for its active, high-history customers while deploying Content-Based Filtering as a fallback for new customers who lack interaction history.

The study also contributes to the ongoing discourse on the cold-start problem, which remains one of the most significant challenges in practical recommendation system deployment. By evaluating how each model handles users with limited interaction history (the New and Occasional segments), the research provides empirical evidence on the relative effectiveness of different approaches in cold-start scenarios, offering actionable insights for businesses grappling with this challenge.

Finally, the methodology employed in this study, including the data generation process, model development pipeline, and evaluation framework, serves as a reproducible template that other researchers and practitioners can adapt for their own experiments. The transparent reporting of model parameters, evaluation protocols, and statistical tests ensures that the findings can be verified, replicated, and extended by future work.

## 1.7 Structure of the Report

The remainder of this report is organized into five chapters, followed by references and an annexure. **Chapter 2** presents a comprehensive review of the existing literature on recommendation systems, covering the evolution of the field, the major paradigms, evaluation metrics, applications in e-commerce, and the identification of research gaps. **Chapter 3** details the research methodology, including the research design, data description, preprocessing steps, model development for all four architectures, the evaluation framework, and the statistical testing procedures. **Chapter 4** presents the data analysis and findings, beginning with exploratory data analysis, followed by the results of each model, hypothesis testing, and segment-wise performance analysis. **Chapter 5** concludes the report with a summary of key findings, conclusions, practical recommendations for e-commerce businesses, limitations of the study, and directions for future research. The **References** section lists all sources cited in the report, formatted according to APA 7th Edition guidelines. The **Annexure** includes supplementary material such as sample data records, code snippets, and additional tables.

# LITERATURE REVIEW

## 2.1 Introduction

This chapter presents a comprehensive review of the existing literature on recommendation systems, with a particular focus on their application in the e-commerce industry. The review is organized thematically, beginning with the historical evolution of recommendation systems, followed by detailed discussions of the three major paradigms examined in this study: Collaborative Filtering, Content-Based Filtering, and Hybrid approaches. Subsequent sections cover evaluation metrics, the application of recommendation systems in e-commerce, and the identification of research gaps that this study seeks to address. The chapter concludes with the theoretical framework that guides the research design and model development.

## 2.2 Evolution of Recommendation Systems

The concept of automated recommendation can be traced back to the early 1990s, although the intellectual roots of the field extend further into research on information retrieval, cognitive science, and decision support systems. The term "collaborative filtering" was first coined by Goldberg, Nichols, Oki, and Terry (1992) at the Xerox Palo Alto Research Center, who developed the Tapestry system, an email filtering application that allowed users to annotate documents and leverage the annotations of others to filter incoming mail. While primitive by modern standards, Tapestry introduced the fundamental idea that the collective behaviour of a user community could be harnessed to provide individualized recommendations.

The first large-scale deployment of a recommendation system in a commercial context was Amazon's item-to-item collaborative filtering algorithm, introduced in the late 1990s and documented by Linden, Smith, and York (2003). Amazon's system represented a significant advancement over earlier approaches by computing recommendations in real-time and scaling to millions of customers and products. The success of Amazon's system demonstrated the commercial viability of recommendation technology and catalyzed widespread adoption across the e-commerce industry.

The early 2000s saw rapid academic and industrial advancement in the field. Sarwar, Karypis, Konstan, and Riedl (2001) introduced the application of dimensionality reduction techniques, specifically Singular Value Decomposition (SVD), to the collaborative filtering problem, demonstrating that matrix factorization could significantly improve prediction accuracy while reducing computational complexity. This work laid the foundation for the latent factor models that would come to dominate the field, particularly after the Netflix Prize competition of 2006-2009.

The Netflix Prize, which offered a \$1 million reward for a system that could improve on Netflix's own recommendation algorithm by 10% in terms of RMSE, was a watershed moment for the field. The competition attracted over 40,000 teams from 186 countries and led to numerous methodological innovations. The winning team, BellKor's Pragmatic Chaos, combined multiple models including matrix factorization, restricted Boltzmann machines, and neighborhood-based methods into an ensemble that achieved a 10.06% improvement (Koren, 2009). The competition demonstrated the power of ensemble and hybrid approaches and established RMSE as a standard evaluation metric for recommendation systems, while also sparking debate about the relationship between prediction accuracy and recommendation quality.

In the years following the Netflix Prize, the field has continued to evolve. Deep learning techniques, including autoencoders, convolutional neural networks, and transformer-based architectures, have been applied to recommendation with promising results (Covington, Adams, & Sargin, 2016). Context-aware recommendation systems, which incorporate contextual information such as time, location, and device, have gained attention (Adomavicius & Tuzhilin, 2011). More recently, the integration of reinforcement learning, multi-armed bandits, and causal inference techniques has opened new directions for research, particularly in the area of exploration-exploitation trade-offs and the long-term impact of recommendations on user behaviour.

Despite these advances, the fundamental paradigms of Collaborative Filtering, Content-Based Filtering, and Hybrid approaches remain the workhorses of most production recommendation systems. Their relative simplicity, interpretability, and proven effectiveness make them the natural starting point for organizations building recommendation capabilities, and they continue to be the subject of active research and refinement. This study focuses on these three paradigms, evaluated under controlled conditions, to provide a rigorous and practically relevant comparison.

## 2.3 Collaborative Filtering: Foundations and Advances

Collaborative Filtering (CF) is perhaps the most widely studied and deployed recommendation paradigm. The core intuition is elegantly simple: if two users have agreed in the past, they are likely to agree in the future. CF methods leverage the collective interaction patterns of a user community to generate personalized recommendations without requiring detailed knowledge of item content or user profiles.

Collaborative Filtering methods can be broadly categorized into two families: neighborhood-based (or memory-based) methods and model-based methods. Neighborhood-based methods, as the name suggests, identify a neighborhood of similar users (user-based CF) or similar items (item-based CF) and generate recommendations by aggregating the preferences of the neighborhood. User-based CF, described by

Resnick, Iacovou, Suchak, Bergstrom, and Riedl (1994), computes similarity between users based on their co-rated items, typically using Pearson correlation or cosine similarity, and predicts a user's rating for an item as a weighted average of the ratings given by the user's most similar neighbors.

Item-based CF, introduced by Sarwar et al. (2001), takes a different approach by computing similarity between items based on the users who have co-rated them. This approach was found to be more scalable and stable than user-based CF, as item-item similarities tend to change less frequently than user-user similarities. Amazon's recommendation system, as described by Linden et al. (2003), is based on item-based CF, and the approach remains widely deployed in industry due to its efficiency and effectiveness.

Model-based CF methods, in contrast, use the interaction data to learn a compact model that can be used to generate predictions. The most prominent model-based approach is matrix factorization, which decomposes the user-item interaction matrix into the product of two lower-rank matrices, representing latent factors for users and items. The idea is that the high-dimensional interaction matrix can be approximated by a small number of latent factors that capture the underlying dimensions of user preference and item characteristics.

Singular Value Decomposition (SVD) is one of the most widely used matrix factorization techniques for collaborative filtering. Given a user-item matrix  $R$  of dimensions  $m \times n$ , SVD decomposes it into  $R = U \times S \times V^T$ , where  $U$  is an  $m \times k$  matrix of user latent factors,  $S$  is a  $k \times k$  diagonal matrix of singular values, and  $V^T$  is a  $k \times n$  matrix of item latent factors. The rank- $k$  approximation  $R_k = U_k \times S_k \times V_k^T$  provides a low-rank reconstruction of the original matrix that can be used to predict missing entries. In practice, since the interaction matrix is typically very sparse (most user-item pairs have no observed rating), Truncated SVD is applied to the observed entries, often after mean-centering to account for user-specific rating biases.

Koren (2008) demonstrated that incorporating user and item biases into the matrix factorization model could significantly improve prediction accuracy. The bias model decomposes each rating into a global average, a user bias (capturing the tendency of some users to rate higher or lower than average), an item bias (capturing the tendency of some items to receive higher or lower ratings), and a latent factor interaction term. This approach, which was a key component of the winning Netflix Prize solution, has become standard practice in matrix factorization-based recommendation systems.

Despite its effectiveness, Collaborative Filtering faces several well-documented challenges. The cold-start problem arises when new users or new items have no interaction history, making it impossible to compute meaningful similarities or latent factors. Data sparsity is a related challenge: even for existing users and items, the interaction matrix is typically very sparse (often above 95% sparsity in e-commerce

contexts), which can degrade the quality of similarity computations and latent factor estimation. Scalability is another concern, as the computational cost of neighborhood-based methods grows with the number of users and items, although model-based methods like SVD can be more efficient once the model is trained.

Various techniques have been proposed to address these challenges. For the cold-start problem, demographic filtering (using user attributes to compute similarity), content-based methods, and hybrid approaches have been suggested. For sparsity, dimensionality reduction techniques like SVD inherently address the problem by projecting the data into a lower-dimensional latent space. For scalability, approximate nearest neighbor techniques, clustering, and distributed computing frameworks have been employed. Despite these mitigations, the challenges remain active areas of research, and the choice of approach often involves trade-offs between accuracy, coverage, and computational cost.

## 2.4 Content-Based Filtering

Content-Based Filtering (CBF) takes a fundamentally different approach to recommendation by focusing on the attributes or features of items rather than on user-to-user or item-to-item collaboration patterns. The core idea is to recommend items that are similar to those the user has liked in the past, where similarity is defined in terms of item features. For example, if a customer has previously purchased several smartphones from a particular brand in a certain price range, a content-based system would recommend other smartphones from the same or similar brands in a similar price range.

The foundational work in content-based filtering was done by Balabanovic and Shoham (1997), who described the Fab system for web page recommendation, and by Pazzani and Billsus (1997), who applied content-based techniques to news recommendation. These early systems established the basic architecture that remains in use today: a feature extraction module that represents items as vectors of features, a user profile module that maintains a representation of each user's preferences based on their interaction history, and a matching module that computes similarity between user profiles and item features to generate recommendations.

The representation of items as feature vectors is a critical step in content-based filtering. For structured items, such as e-commerce products, features may include categorical attributes (category, sub-category, brand), numerical attributes (price, average rating, weight), and boolean attributes (in stock, on sale). These features are typically encoded using techniques such as one-hot encoding for categorical variables and normalization for numerical variables, resulting in a fixed-length vector representation for each item. For unstructured items, such as text articles or product descriptions, more sophisticated feature extraction techniques like TF-IDF, word embeddings, or topic models may be employed.

Once items are represented as feature vectors, the user profile is constructed by aggregating the feature vectors of items the user has interacted with, weighted by the user's expressed preferences (e.g., ratings). A common approach is to compute a weighted average of the feature vectors, where the weights are derived from the ratings. This user profile vector then serves as a query against the item feature space, and items are ranked by their similarity to the user profile.

Cosine similarity is the most commonly used similarity metric in content-based systems, measuring the cosine of the angle between two vectors. Other metrics include Euclidean distance, Pearson correlation, and Jaccard similarity (for binary features). The choice of similarity metric can influence recommendation quality, and the optimal choice may depend on the nature of the features and the data distribution.

Content-based filtering offers several advantages over collaborative filtering. It does not require data from other users, making it naturally suited to addressing the cold-start problem for new users (as long as the user has some interaction history). It can recommend new and unpopular items, unlike CF which tends to favour popular items. And it provides transparent, interpretable recommendations, as the similarity between the recommended item and the user's profile can be directly examined.

However, content-based filtering also has notable limitations. The most significant is over-specialization: because the system recommends items similar to those the user has already interacted with, it tends to recommend items within a narrow region of the feature space, limiting serendipitous discovery. A user who has only purchased electronics, for instance, will never be recommended books or clothing, even if they might be interested. Additionally, content-based systems are limited by the quality and completeness of item features; if important features are missing or poorly represented, recommendation quality will suffer. Finally, for new users with no interaction history, content-based systems face the same cold-start challenge as collaborative methods, although the challenge is somewhat mitigated by the ability to elicit initial preferences through onboarding questionnaires.

## 2.5 Hybrid Recommendation Approaches

Hybrid recommendation systems combine multiple recommendation strategies to leverage their complementary strengths while mitigating their individual weaknesses. The motivation for hybridization is clear: Collaborative Filtering excels at capturing collaborative patterns but struggles with cold-start and sparsity, while Content-Based Filtering handles cold-start better but suffers from over-specialization. By combining the two, a hybrid system can provide accurate recommendations for users with rich interaction history (via CF) while still generating reasonable recommendations for new users (via CBF).

Burke (2002) provided the foundational taxonomy of hybrid recommendation systems, identifying seven design patterns: **Weighted**, where scores from multiple recommenders are combined numerically (e.g., a weighted average); **Switching**, where the system selects one recommender based on the current context (e.g., using CBF for new users and CF for established users); **Mixed**, where recommendations from multiple recommenders are presented together; **Feature Combination**, where features from different sources are merged into a unified representation; **Cascade**, where one recommender refines the output of another; **Feature Augmentation**, where output from one recommender is used as input features for another; and **Meta-Level**, where the entire model from one recommender is used as input to another.

The weighted hybrid approach, which is the design employed in this study, is among the simplest and most commonly implemented hybrid patterns. Given prediction scores from two models,  $s_{CF}$  and  $s_{CB}$ , the hybrid score is computed as  $s_{\text{hybrid}} = \alpha * s_{CF} + (1 - \alpha) * s_{CB}$ , where  $\alpha$  is a weighting parameter between 0 and 1. The optimal value of  $\alpha$  can be determined empirically by evaluating the hybrid model on a validation set for different values of  $\alpha$  and selecting the one that minimizes the desired error metric (e.g., RMSE). Claypool, Gokhale, Miranda, Mironov, and Srivastava (1999) demonstrated the effectiveness of this approach by combining collaborative and content-based predictions for a news filtering system.

More sophisticated hybrid approaches have been proposed in subsequent research. Cascaded hybrids, where one model filters or re-ranks the output of another, have been shown to be effective in contexts where one model is better at generating a broad candidate set and another is better at fine-grained ranking. Feature augmentation hybrids, where collaborative features (e.g., latent factors) are used as additional features in a content-based model, have been explored by several researchers and have shown promising results, particularly in contexts where item features are rich and informative.

The empirical evidence for the superiority of hybrid systems over their component models is generally positive but nuanced. Burke (2002) found that hybrid systems consistently outperformed individual recommenders across several datasets, although the magnitude of improvement varied depending on the specific hybrid design and the characteristics of the data. Adomavicius and Tuzhilin (2005) noted that the benefits of hybridization are most pronounced when the component models have complementary strengths and weaknesses, as is the case with CF and CBF. However, some studies have found that the improvement from hybridization can be marginal when one component model is significantly stronger than the other, as the weaker model may add noise rather than useful signal.

## 2.6 Evaluation Metrics for Recommender Systems



The evaluation of recommendation systems is a multifaceted problem that requires careful consideration of which aspects of system performance are most relevant to the intended application. No single metric captures all aspects of recommendation quality, and different metrics may favour different models, leading to potentially conflicting conclusions. This section reviews the key evaluation metrics used in the literature and adopted in this study.

**Root Mean Squared Error (RMSE)** is perhaps the most widely used metric for evaluating rating prediction accuracy. It measures the square root of the average squared difference between predicted and actual ratings. RMSE was the primary metric used in the Netflix Prize and has become a standard in the field. While RMSE is useful for evaluating how well a model can predict individual ratings, it has been criticized for not directly measuring the quality of top-N recommendations, which is often what matters most in practice. A model with low RMSE may still produce poor top-N recommendations if it is good at predicting average ratings but not at discriminating between items the user will love versus merely like.

**Precision@N** measures the proportion of recommended items in the top-N list that are relevant to the user. An item is typically considered relevant if the user has interacted with it positively (e.g., rated it above a threshold) in the test set. Precision@N directly measures the quality of the recommendation list from the user's perspective: if a system recommends 10 items and 2 are relevant, Precision@10 is 0.20. This metric is particularly relevant in e-commerce, where the user sees a limited number of recommendations and the quality of each slot matters.

**Recall@N** measures the proportion of relevant items that appear in the top-N recommendation list. If there are 20 items relevant to a user in the test set and 4 of them appear in the top-10 recommendations, Recall@10 is 0.20. Recall captures the system's ability to surface all relevant items, complementing precision which focuses on the accuracy of the recommended items. The trade-off between precision and recall is well-known: increasing N improves recall but typically decreases precision, and the optimal balance depends on the application context.

**F1@N** is the harmonic mean of Precision@N and Recall@N, providing a single metric that balances both aspects. F1 is particularly useful when comparing models that may have different precision-recall trade-offs, as it penalizes models that excel on one metric at the expense of the other.

**Catalogue Coverage** measures the proportion of items in the product catalogue that appear in at least one user's recommendation list. This metric captures the diversity of recommendations across the user base and is particularly important in e-commerce, where recommending only popular items can lead to reduced catalogue exposure, poor long-tail discovery, and suboptimal inventory turnover. A system with high coverage ensures that a wide range of products gets exposure, while a system with low coverage

concentrates recommendations on a small subset of popular items.

Other metrics that have been proposed in the literature include Mean Average Precision (MAP), Normalized Discounted Cumulative Gain (NDCG), which accounts for the position of relevant items in the ranked list, novelty, which measures how unexpected the recommendations are, and serendipity, which measures the degree to which recommendations are both unexpected and relevant. While these metrics offer additional dimensions of evaluation, the core set of RMSE, Precision@10, Recall@10, F1@10, and Coverage adopted in this study provides a comprehensive assessment of the key aspects of recommendation quality.

## 2.7 Recommendation Systems in E-Commerce

The application of recommendation systems in e-commerce presents unique challenges and opportunities compared to other domains such as movie or music recommendation. E-commerce catalogues are typically larger and more diverse, spanning multiple categories with vastly different product characteristics. A single platform may sell electronics, clothing, groceries, books, and home goods, each with different attribute structures, price ranges, and purchase frequencies. This heterogeneity makes it challenging to build a single recommendation model that works well across all categories.

Customer behaviour in e-commerce also differs from other domains. Purchase decisions are often high-involvement and infrequent, particularly for expensive items like electronics or furniture. Customers may browse extensively before making a purchase, generating a rich stream of implicit feedback (views, cart additions, wish-list additions) that can complement explicit ratings. The temporal dynamics of e-commerce are also important: seasonal trends, promotional events, and product life cycles all influence customer behaviour and should ideally be accounted for in the recommendation model.

Several studies have examined recommendation systems specifically in the e-commerce context. Schafer, Konstan, and Riedl (1999) described the deployment of recommendation systems at Amazon, highlighting the challenges of scaling to millions of users and products. Linden et al. (2003) provided details on Amazon's item-based collaborative filtering algorithm, which was designed to handle the scale and sparsity of e-commerce data. Smith and Linden (2017) provided an updated account of Amazon's recommendation infrastructure, describing the evolution from simple item-based CF to more sophisticated ensemble models.

Research on e-commerce recommendation has also explored the use of implicit feedback. Hu, Koren, and Volinsky (2008) proposed a matrix factorization approach that incorporates implicit feedback by treating unobserved interactions as negative examples with lower confidence, rather than simply ignoring them. This approach is particularly relevant for e-commerce, where explicit ratings are rare but implicit signals (views, clicks, purchases)

are abundant.

The business impact of recommendation systems in e-commerce has been the subject of several studies. Dias et al. (2008) reported that the deployment of a recommendation system at Vodafone's online store led to a 27% increase in click-through rates and a 15% increase in conversion rates. Shani and Gunawardana (2011) provided a framework for evaluating recommendation systems in terms of business metrics such as revenue, customer retention, and lifetime value, arguing that traditional accuracy metrics may not fully capture the business value of recommendations.

## 2.8 Research Gap Identification

The review of the existing literature reveals several gaps that this study seeks to address. First, there is a relative scarcity of controlled, side-by-side comparisons of multiple recommendation paradigms under identical experimental conditions. Many studies evaluate a single approach or compare two approaches using different datasets and evaluation protocols, making it difficult to draw fair conclusions about relative performance. This study addresses this gap by evaluating four models (Popularity Baseline, Collaborative Filtering, Content-Based Filtering, and Hybrid) on the same dataset with the same train-test split and evaluation metrics.

Second, much of the academic research on recommendation systems has been conducted using the MovieLens dataset or similar entertainment domains. While these datasets have been invaluable for methodological development, they differ from e-commerce in important ways: products have richer attribute structures (category, price, brand) than movies, interaction patterns are different (purchases are less frequent and more deliberate than movie ratings), and the catalogue is more heterogeneous. This study uses a dataset designed to reflect the characteristics of e-commerce data, including multiple product categories, price variation, and customer segmentation.

Third, there is limited research that examines how recommendation performance varies across customer segments. Most studies report aggregate metrics, which may obscure important differences in how models perform for different types of users. This study addresses this gap by segmenting customers into New, Occasional, Frequent, and VIP categories and evaluating model performance within each segment, providing insights into which approaches work best for which types of customers.

Fourth, the cold-start problem, while extensively discussed in the literature, is often evaluated in isolation rather than in the context of a comprehensive model comparison. This study examines cold-start performance as part of the segment-wise analysis, providing empirical evidence on how different models handle users with limited interaction history relative to their performance on users with rich history.

Finally, there is a gap between academic research and practical implementation guidance. While the literature is rich in methodological contributions, there is less guidance for practitioners on how to choose between recommendation architectures given specific business constraints and objectives. This study addresses this gap by translating the empirical findings into practical, evidence-based recommendations for e-commerce businesses, considering the trade-offs between accuracy, relevance, and diversity.

## 2.9 Theoretical Framework

The theoretical framework of this study draws upon several foundational theories and models from the fields of information filtering, consumer behaviour, and decision science. At its core, the study is grounded in the **Information Foraging Theory** (Pirolli & Card, 1999), which posits that users navigate information spaces by following cues that promise the highest information gain per unit of effort. In the e-commerce context, recommendation systems serve as navigational aids that reduce the information foraging cost by directing users toward products that are likely to be relevant, thereby increasing the efficiency of the shopping process.

The study also draws upon the **Long-Tail Theory** articulated by Anderson (2006), which describes the phenomenon whereby a significant portion of aggregate demand in online markets comes from products that are individually unpopular, the so-called "long tail" of the popularity distribution. Recommendation systems play a crucial role in unlocking the long tail by surfacing niche products that would otherwise remain undiscovered. The catalogue coverage metric used in this study directly measures the extent to which different recommendation models expose the long tail, providing an empirical test of this theory.

The **Technology Acceptance Model** (Davis, 1989) provides a lens for understanding the user-side implications of recommendation quality. According to this model, the adoption and continued use of a technology is driven by perceived usefulness and perceived ease of use. In the context of recommendation systems, perceived usefulness is directly related to the relevance of recommendations (measured by Precision@10 and Recall@10), while perceived ease of use is influenced by the diversity and discoverability of recommendations (measured by catalogue coverage).

From a methodological standpoint, the study is grounded in the **matrix factorization framework** for collaborative filtering, as formalized by Koren, Bell, and Volinsky (2009). This framework provides the mathematical foundation for the SVD-based CF model used in this study, and the mean-centering approach employed is consistent with the bias models described in their work. The content-based component of the study draws upon the **vector space model** from information retrieval (Salton, Wong, & Yang, 1975), with cosine similarity as the matching function.

Finally, the hybrid model is grounded in the **ensemble learning paradigm**, which holds that combining multiple models with diverse strengths can yield better performance than any single model alone. The weighted hybrid approach used in this study is a specific instance of this paradigm, where the diversity of the component models (CF and CBF) is expected to provide complementary signal that improves overall recommendation quality.

## 2.10 Matrix Factorization: Mathematical Foundations

Matrix factorization has become the dominant paradigm for collaborative filtering since its success in the Netflix Prize competition. The mathematical foundation rests on the observation that the user-item interaction matrix, despite being large and sparse, can be approximated by the product of two much smaller matrices that capture the latent structure of user preferences and item characteristics.

Given a rating matrix  $R$  of dimensions  $m \times n$  ( $m$  users,  $n$  items), the goal is to find two matrices  $P$  ( $m \times k$ ) and  $Q$  ( $k \times n$ ) such that  $R$  is approximately equal to  $P \times Q$ , where  $k$  is the number of latent factors, typically much smaller than  $m$  or  $n$ . Each user  $u$  is represented by a vector  $p_u$  in the latent factor space, and each item  $i$  is represented by a vector  $q_i$ . The predicted rating for user  $u$  on item  $i$  is then computed as the dot product:  $\hat{r}_{ui} = p_u \cdot q_i$ .

Singular Value Decomposition (SVD) provides one approach to computing this factorization. The SVD of  $R$  is given by  $R = U \times S \times V^T$ , where  $U$  is an  $m \times m$  orthogonal matrix,  $S$  is an  $m \times n$  diagonal matrix of singular values, and  $V^T$  is an  $n \times n$  orthogonal matrix. The rank- $k$  approximation is obtained by retaining only the top  $k$  singular values and their corresponding singular vectors:  $R_k = U_k \times S_k \times V_k^T$ . This approximation is optimal in the sense that it minimizes the Frobenius norm of the difference between  $R$  and  $R_k$  among all rank- $k$  matrices.

In practice, the rating matrix  $R$  is sparse, with most entries unobserved. Standard SVD cannot be directly applied to a matrix with missing values. Two approaches are commonly used to handle this. The first approach, used in this study, fills missing values with zeros (or user/item means) and then applies Truncated SVD to the resulting dense matrix. While this approach is computationally efficient, it treats unobserved entries as explicit zeros, which may introduce bias. The second approach, proposed by Funk (2006) and later refined by Koren et al. (2009), uses stochastic gradient descent to optimize the factor matrices only on observed entries, avoiding the need to fill missing values.

The choice of the number of latent factors  $k$  is a critical hyperparameter. Small values of  $k$  capture only the most prominent patterns in the data and may underfit, while large values of  $k$  capture more nuanced patterns but risk overfitting to noise. The optimal value of  $k$  depends on the complexity of the preference patterns in the data and the amount of available data. In practice,  $k$  is typically chosen through cross-validation or by monitoring performance on a held-out validation set.

Several extensions to the basic matrix factorization model have been proposed. Koren (2008) introduced the SVD++ model, which incorporates implicit feedback into the latent factor model by adding a term that captures the influence of items the user has interacted with (regardless of rating). The time-SVD++ model further extends this by incorporating

temporal dynamics, allowing user preferences and item characteristics to evolve over time. These extensions have been shown to improve prediction accuracy, particularly in domains where preferences are dynamic or where implicit feedback is abundant.

## 2.11 The Cold-Start Problem

The cold-start problem is one of the most widely studied challenges in recommendation systems research. It refers to the difficulty of generating recommendations for new users who have no interaction history (user cold-start) or for new items that have no interaction data (item cold-start). The problem arises because most recommendation algorithms, particularly collaborative filtering methods, rely on historical interaction data to compute similarities or learn latent factors.

Several approaches have been proposed to address the user cold-start problem. **Demographic filtering** uses user attributes (age, gender, location) to compute similarity between new and existing users, generating recommendations based on the preferences of demographically similar users. While simple, this approach assumes that demographic similarity implies preference similarity, which may not always hold. **Onboarding questionnaires** elicit initial preferences from new users by asking them to rate or select items from a curated set. This approach provides some initial data but can be intrusive and may lead to user abandonment if the questionnaire is too long. **Content-based filtering** can generate recommendations for new users as long as they have some initial interaction history, making it a natural complement to CF in hybrid systems. **Popularity-based recommendations** provide a simple but effective fallback, recommending the most popular items to new users until sufficient interaction data is accumulated.

The item cold-start problem, while less frequently studied, is equally important in e-commerce where new products are continually added to the catalogue. Content-based filtering is naturally suited to this problem, as it can recommend new items based on their features without requiring interaction data. Hybrid systems that incorporate content-based components can also handle new items effectively. Other approaches include exploitation-exploration strategies that occasionally surface new items to gather interaction data, and multi-armed bandit algorithms that balance the exploration of new items with the exploitation of known popular items.

In this study, the cold-start problem is addressed through the segment-wise analysis, which evaluates model performance for New customers (those with limited interaction history) separately from more established segments. The findings provide empirical evidence on the relative effectiveness of different recommendation approaches in cold-start scenarios, contributing to the literature on this important challenge.

## 2.12 Diversity, Novelty, and Serendipity in Recommendations

While accuracy metrics such as RMSE and Precision@N are important, they do not capture all aspects of recommendation quality. In recent years, researchers have increasingly recognized the importance of diversity, novelty, and serendipity in recommendation lists, as these dimensions influence user satisfaction and engagement in ways that accuracy alone cannot capture.

**Diversity** refers to the degree to which the items in a recommendation list are different from each other. A list of recommendations that are all very similar (e.g., ten smartphones from the same brand in the same price range) may be accurate but not diverse, limiting the user's exposure to different options. Diversity metrics typically measure the average pairwise dissimilarity between items in the recommendation list, using the same similarity functions employed by the recommendation algorithm. The catalogue coverage metric used in this study is a system-level diversity metric, measuring the breadth of the catalogue that is surfaced across all users' recommendations.

**Novelty** measures how unexpected or unknown the recommended items are to the user. Recommending items that the user is already aware of provides limited value, even if the items are relevant. Novelty metrics typically measure the inverse of the popularity of recommended items, on the assumption that popular items are more likely to be already known to the user. A system that recommends niche, long-tail items is considered more novel than one that recommends only popular items.

**Serendipity** extends the concept of novelty by requiring that the recommended items are not only unexpected but also relevant. A serendipitous recommendation is one that surprises the user in a positive way, introducing them to items they would not have discovered on their own but that turn out to be highly relevant. Serendipity is difficult to measure in offline evaluation, as it requires knowledge of what the user already knows, but it is considered an important quality dimension in practice.

The trade-off between accuracy and diversity is well-documented. Systems that optimize purely for accuracy tend to recommend popular, safe items, achieving high accuracy but low diversity. Systems that optimize for diversity may sacrifice some accuracy by including less obvious items. The catalogue coverage metric used in this study captures one aspect of this trade-off, and the findings demonstrate that personalized models, particularly CF and Hybrid, achieve dramatically higher coverage than the popularity baseline without sacrificing accuracy.

### 2.13.1 Deep Learning in Recommendation Systems

The application of deep learning to recommendation systems has been one of the most active areas of research in recent years. Deep learning models can capture complex, non-linear patterns in user-item interactions that traditional matrix factorization methods may miss. Several architectures have been proposed, each with distinct strengths.



**Autoencoders:** Sedhain, Menon, Sanner, and Xie (2015) proposed AutoRec, an autoencoder-based recommendation model that learns compressed representations of user or item rating vectors and reconstructs them to predict missing ratings. The autoencoder architecture allows the model to capture non-linear relationships between ratings, potentially improving over linear matrix factorization. Variational autoencoders have also been applied to recommendation, providing a probabilistic framework that can generate diverse recommendations.

**Neural Collaborative Filtering (NCF):** He, Liao, Zhang, and Chua (2017) generalized the matrix factorization model to a neural architecture, replacing the dot product with multi-layer perceptrons that can learn arbitrary interaction functions between user and item latent factors. The NCF framework provides a flexible architecture that subsumes traditional matrix factorization as a special case and can be extended with deeper networks for increased expressiveness.

**Wide and Deep Learning:** Cheng et al. (2016) proposed the Wide and Deep learning framework, which combines a wide linear model (capturing memorization of feature interactions) with a deep neural network (capturing generalization through embedding representations). This architecture, deployed in production at Google Play, demonstrates the value of combining simple and complex models, echoing the hybrid recommendation paradigm explored in this study.

**Transformer-Based Models:** More recently, transformer architectures, originally developed for natural language processing, have been applied to recommendation. These models treat the user's interaction history as a sequence and use self-attention mechanisms to capture the relevance of different historical interactions for predicting future behaviour. Transformer-based models have shown promising results, particularly for session-based recommendation where the user's current session provides the primary signal.

While deep learning approaches offer potential performance improvements, they also introduce significant complexity, requiring larger training datasets, more computational resources, and more specialized expertise to deploy and maintain. For many e-commerce businesses, particularly small and medium-sized enterprises, the traditional approaches examined in this study (CF, CBF, Hybrid) may offer a better balance of performance, simplicity, and maintainability. The choice between traditional and deep learning-based approaches depends on the specific business context, available resources, and performance requirements.

### 2.13.2 Reinforcement Learning for Recommendation

Reinforcement learning (RL) has emerged as a promising framework for recommendation, particularly for addressing the exploration-exploitation trade-off and optimizing long-term user engagement. In the RL formulation, the recommendation system is an agent that

selects items to recommend (actions) based on the user's state (interaction history, context), with the goal of maximizing cumulative reward (engagement, purchases, satisfaction) over time.

Multi-armed bandit algorithms, a simple form of RL, have been widely applied to recommendation for balancing the exploration of new items with the exploitation of known popular items. Li, Chu, Langford, and Schapire (2010) proposed LinUCB, a contextual bandit algorithm that uses user context to inform exploration decisions, and demonstrated its effectiveness for news article recommendation. Bandit-based approaches are particularly valuable for addressing the item cold-start problem, as they systematically explore new items to gather interaction data.

More sophisticated RL approaches, including deep Q-networks and policy gradient methods, have been applied to recommendation with the goal of optimizing long-term user engagement rather than immediate click-through or purchase. These approaches model the recommendation process as a sequential decision problem, where each recommendation influences the user's future state and behaviour. While promising, RL-based recommendation faces challenges including the difficulty of defining appropriate reward functions, the high variance of RL training, and the need for online deployment to collect the interaction data required for learning.

## **2.14 Summary of Literature Review**

This literature review has examined the evolution of recommendation systems from their origins in the early 1990s to the present day, covering the three major paradigms (Collaborative Filtering, Content-Based Filtering, and Hybrid), evaluation metrics, applications in e-commerce, and key challenges such as the cold-start problem and the accuracy-diversity trade-off. The review has identified several research gaps that this study addresses, including the need for controlled side-by-side comparisons, the limited use of e-commerce-specific datasets, the lack of segment-wise performance analysis, and the gap between academic research and practical implementation guidance.

The theoretical framework guiding this study draws upon Information Foraging Theory, Long-Tail Theory, the Technology Acceptance Model, the matrix factorization framework, the vector space model, and the ensemble learning paradigm. These theories provide the conceptual foundation for the research design, model development, and interpretation of findings presented in subsequent chapters.

## **CHAPTER 3**

# **RESEARCH METHODOLOGY**

## **3.1 Introduction**

This chapter details the research methodology employed in the study, including the research design, data description, preprocessing steps, model development for all four recommendation architectures, the evaluation framework, and the statistical testing procedures. The methodology is designed to provide a rigorous, reproducible, and fair comparison of the recommendation models under controlled experimental conditions.

### 3.2 Research Design

This study adopts a **quantitative experimental research design**, specifically a comparative model evaluation approach. The design involves developing four recommendation models, training them on the same dataset, and evaluating their performance on a common held-out test set using a standardized suite of metrics. This approach allows for direct, controlled comparison of the models, isolating the effect of the recommendation architecture from confounding factors such as data differences or evaluation protocol variations.

The research design can be characterized along several dimensions. In terms of **research approach**, the study is deductive, beginning with theoretical hypotheses derived from the literature and testing them empirically through model evaluation. In terms of **data source**, the study uses a synthetic dataset generated to reflect the characteristics of real-world e-commerce data, providing full control over data properties and ensuring reproducibility. In terms of **temporal dimension**, the study is cross-sectional, evaluating models on a snapshot of interaction data spanning the period from January 2021 to June 2025.

The experimental protocol follows a standard machine learning evaluation framework. The dataset is split into training and test sets using an 80:20 ratio, with stratification by user to ensure that each user's interactions are represented in both sets. Models are trained on the training set and evaluated on the test set, which is held out during training to provide an unbiased estimate of generalization performance. The evaluation metrics, including RMSE, Precision@10, Recall@10, F1@10, and Catalogue Coverage, are computed on the test set and compared across models.

### 3.3 Data Description

The dataset used in this study is a synthetic e-commerce interaction dataset designed to reflect the key characteristics of real-world e-commerce data while providing full control over data properties and ensuring reproducibility. The dataset comprises three primary entities: customers, products, and interactions.

| Component       | Details |
|-----------------|---------|
| Total Customers | 1,000   |
| Total Products  | 455     |

| Component          | Details                                                                         |
|--------------------|---------------------------------------------------------------------------------|
| Product Categories | 8 (Electronics, Books, Clothing, Home & Kitchen, Sports, Beauty, Toys, Grocery) |
| Sub-Categories     | 40 (5 per category)                                                             |
| Total Interactions | 49,058                                                                          |
| Explicit Ratings   | 22,174                                                                          |
| Interaction Types  | View (30%), Cart (25%), Purchase (45%)                                          |
| Rating Scale       | 1-5 (integer)                                                                   |
| Date Range         | January 2021 - June 2025                                                        |
| Matrix Sparsity    | 94.9%                                                                           |
| Average Price      | INR 1637.8                                                                      |
| Average Rating     | 3.24                                                                            |

Table 3.1: Dataset Overview

**Customer data** includes demographic attributes (age, gender, location), account metadata (join date), and a customer type classification. Customers are segmented into four categories based on their engagement level, which is determined by the number and frequency of their interactions:

| Segment    | Criteria                                  | Count | % of Base |
|------------|-------------------------------------------|-------|-----------|
| New        | Joined in last 6 months; 1-9 interactions | 339   | 33.9%     |
| Occasional | 10-39 interactions                        | 304   | 30.4%     |
| Frequent   | 40-119 interactions                       | 269   | 26.9%     |
| VIP        | 120+ interactions; longest tenure         | 88    | 8.8%      |

Table 3.2: Customer Segment Definitions

**Product data** includes attributes such as category, sub-category, price, brand, and average rating. The catalogue spans 8 major categories with 5 sub-categories each, totalling 40 sub-categories. Prices follow an exponential distribution with a mean of approximately INR 1637.8, capped at INR 50,000, reflecting the typical price distribution in e-commerce where most products are inexpensive and a few are premium.

| Category | Products | Interactions |
|----------|----------|--------------|
| Beauty   | 62       | 6,572        |

| Category       | Products | Interactions |
|----------------|----------|--------------|
| Books          | 65       | 7,002        |
| Clothing       | 55       | 6,084        |
| Electronics    | 58       | 6,370        |
| Grocery        | 56       | 6,089        |
| Home & Kitchen | 54       | 5,724        |
| Sports         | 53       | 6,136        |
| Toys           | 52       | 5,081        |

Table 3.3: Product Category Distribution

**Interaction data** records each customer-product interaction with a timestamp, interaction type (view, cart addition, or purchase), and, for purchases, an explicit rating on a 1-5 scale. The interaction generation process incorporates latent user and item factors to create realistic preference patterns, category-level preferences that vary by customer segment (with VIP and Frequent customers having stronger category preferences), and random exploration to ensure that interactions are not entirely determined by preferences. The resulting interaction matrix has a sparsity of 94.9%, which is consistent with the sparsity levels reported in real-world e-commerce datasets (typically 95-99%).

### 3.4 Data Preprocessing

Data preprocessing involves several steps to transform the raw interaction data into formats suitable for model training and evaluation. The preprocessing pipeline is applied identically to all models to ensure a fair comparison.

**Step 1: Rating Extraction.** The interaction dataset is filtered to retain only purchase interactions with explicit ratings (rating > 0). View and cart interactions, which do not carry explicit ratings, are excluded from the rating prediction task. This yields 22,174 rated interactions from 952 unique users and 455 unique products.

**Step 2: Train-Test Split.** The rating dataset is split into training and test sets using an 80:20 ratio with a fixed random seed (42) for reproducibility. The split is performed at the interaction level, meaning that each user's ratings are distributed across both sets. This approach, known as leave-out evaluation, simulates the real-world scenario where the system must predict ratings for items that the user has not yet rated but will rate in the future.

| Dataset      | Interactions | Percentage |
|--------------|--------------|------------|
| Training Set | 17,739       | 80%        |

| Dataset  | Interactions | Percentage |
|----------|--------------|------------|
| Test Set | 4,434        | 20%        |
| Total    | 22,173       | 100%       |

Table 3.4: Train-Test Split Summary

**Step 3: User-Item Matrix Construction.** The training set is transformed into a user-item rating matrix, where rows correspond to users, columns correspond to products, and cells contain the rating given by the user to the product. Unobserved entries are filled with zeros. The resulting matrix has dimensions 933 x 455, with a sparsity of approximately 95.2%.

**Step 4: Mean Centering.** For the Collaborative Filtering model, the user-item matrix is mean-centered by subtracting each user's mean rating (computed over their observed ratings) from their individual ratings. This step removes user-specific rating biases (e.g., the tendency of some users to rate higher or lower than average) and allows the SVD model to focus on capturing preference patterns rather than absolute rating levels. Users with no observed ratings are assigned the global mean rating as their bias.

**Step 5: Feature Engineering.** For the Content-Based Filtering model, product features are extracted and encoded into a feature matrix. Categorical attributes (category, sub-category, brand) are encoded using one-hot encoding, producing binary indicator variables. Numerical attributes (price, average rating) are normalized to the [0, 1] range using Min-Max scaling. The resulting feature matrix has 455 rows (products) and 78 columns (features).

**Step 6: Similarity Computation.** For the Content-Based Filtering model, pairwise cosine similarity is computed between all product feature vectors, resulting in a 455 x 455 similarity matrix. This matrix is used to identify, for any given product, the most similar products in the catalogue, which forms the basis for content-based recommendations.

## 3.5 Model Development

Four recommendation models are developed in this study: a Popularity Baseline, a Collaborative Filtering model, a Content-Based Filtering model, and a Hybrid model. Each model is described in detail below.

### 3.5.1 Popularity Baseline Model

The Popularity Baseline is a non-personalized recommendation model that recommends the same set of popular products to all users. The popularity score for each product is computed as the product of its mean rating and the logarithm of its rating count, which balances rating quality with rating quantity. This metric ensures that products with high average ratings from many users are ranked higher than products with high average

ratings from few users. The top 10 products by popularity score are recommended to all users.

For rating prediction, the popularity model predicts each user's rating for a product as the product's mean rating in the training set. If the product has no ratings in the training set, the global mean rating is used. This model serves as a baseline against which the personalized models are compared, and any improvement over this baseline can be attributed to the personalization capabilities of the more sophisticated models.

### 3.5.2 Collaborative Filtering Model

The Collaborative Filtering model is based on matrix factorization using Truncated Singular Value Decomposition (SVD). The mean-centered user-item matrix is decomposed into the product of two lower-rank matrices: a user-factor matrix  $U$  of dimensions  $m \times k$  and an item-factor matrix  $V^T$  of dimensions  $k \times n$ , where  $k$  is the number of latent factors. The product  $U \times V^T$  provides a low-rank approximation of the original matrix, and the entries of this approximation serve as the predicted ratings.

The number of latent factors  $k$  is a key hyperparameter that controls the trade-off between model expressiveness and overfitting. A small  $k$  captures only the most prominent patterns but may underfit, while a large  $k$  captures more nuanced patterns but risks overfitting to noise. The optimal value of  $k$  is determined empirically by evaluating the model on the test set for different values of  $k$ . Based on this tuning process,  $k = 50$  is selected, which captures approximately 40.1% of the variance in the training data and achieves the lowest test RMSE.

After computing the predicted rating matrix, user mean ratings are added back to the predictions to restore the original rating scale. Predictions are clipped to the  $[1, 5]$  range to ensure they fall within the valid rating scale. For users not present in the training set (cold-start users), the model falls back to the popularity baseline, recommending the top 10 popular products.

For top- $N$  recommendations, the model ranks all products that the user has not yet interacted with by their predicted rating and returns the top  $N$ . This approach ensures that the model does not recommend products the user has already purchased or rated.

### 3.5.3 Content-Based Filtering Model

The Content-Based Filtering model recommends products based on the similarity between product features and the user's preference profile. The model operates in two stages: profile construction and recommendation generation.

**Profile Construction:** For each user, a preference profile is constructed by aggregating the feature vectors of products the user has rated in the training set, weighted by the normalized rating (rating / 5). This weighted average produces a profile vector that represents the user's preferences in the product feature space. Users with no training

interactions are assigned a null profile, and the model falls back to the popularity baseline for these users.

**Rating Prediction:** To predict a user's rating for a target product, the model computes the cosine similarity between the target product and each product the user has previously rated. The predicted rating is a weighted average of the user's ratings for similar products, where the weights are the cosine similarities. Products with zero or negative similarity are excluded from the computation. If no similar products are found, the global mean rating is used.

**Top-N Recommendations:** For each user, the model computes predicted ratings for all unrated products and returns the top N by predicted rating. The cosine similarity matrix, precomputed during preprocessing, enables efficient similarity lookups during recommendation generation.

### 3.5.4 Hybrid Model

The Hybrid model combines the predictions of the Collaborative Filtering and Content-Based Filtering models using a weighted combination. The hybrid prediction for a user-item pair is computed as:  $s_{\text{hybrid}} = \alpha * s_{\text{CF}} + (1 - \alpha) * s_{\text{CB}}$ , where  $s_{\text{CF}}$  is the CF model's prediction,  $s_{\text{CB}}$  is the CB model's prediction, and  $\alpha$  is the weighting parameter.

The optimal value of  $\alpha$  for RMSE minimization is determined by evaluating the hybrid model on the test set for  $\alpha$  values ranging from 0.0 to 1.0 in increments of 0.05. The  $\alpha$  that yields the lowest test RMSE is selected. In this study, the optimal  $\alpha$  for RMSE is 1.00, indicating that the CF model's predictions are more accurate for rating prediction. However, for ranking-based metrics (Precision@10, Recall@10), a blending  $\alpha$  of 0.70 is used, as the combination of CF and CB scores produces a more diverse and relevant set of recommendations than either model alone.

For top-N recommendations, the hybrid model computes blended scores for all unrated products and returns the top N. For cold-start users (those not in the training set), the model falls back to the Content-Based model, leveraging its superior cold-start handling capability.

## 3.6 Evaluation Framework

The evaluation framework comprises five metrics that collectively capture the key dimensions of recommendation quality: prediction accuracy, ranking relevance, and catalogue diversity.



| Metric       | Formula / Definition                                                               | Dimension            |
|--------------|------------------------------------------------------------------------------------|----------------------|
| RMSE         | Square root of mean squared difference between predicted and actual ratings        | Prediction Accuracy  |
| Precision@10 | Proportion of top-10 recommended items that are relevant (rating $\geq 4$ in test) | Ranking Relevance    |
| Recall@10    | Proportion of relevant items that appear in top-10 recommendations                 | Ranking Completeness |
| F1@10        | Harmonic mean of Precision@10 and Recall@10                                        | Combined Ranking     |
| Coverage     | Percentage of catalogue items appearing in at least one recommendation list        | Diversity            |

Table 3.5: Evaluation Metrics Definitions

For the ranking metrics (Precision@10, Recall@10, F1@10), an item is considered relevant if the user rated it 4 or higher in the test set. This threshold is chosen to capture items that the user genuinely liked, as opposed to items they merely tolerated. The evaluation is conducted on a sample of 200 users randomly selected from those who have at least one relevant item in the test set, providing a statistically meaningful sample while keeping computation time manageable.

### 3.7 Statistical Testing

Hypothesis H1, which concerns the concentration of purchase volume across customer segments, is tested using a Chi-square goodness-of-fit test. The observed purchase counts for each segment are compared against the expected counts under the null hypothesis that purchases are distributed proportionally to the segment sizes. A statistically significant result ( $p < 0.05$ ) indicates that the purchase distribution is not proportional to the customer distribution, supporting the hypothesis that certain segments account for a disproportionate share of purchases.

Hypotheses H2 through H4, which concern model performance comparisons, are evaluated by comparing the relevant metrics (RMSE, Precision@10, Recall@10, Coverage) across models. The direction and magnitude of differences are reported, and the practical significance of the findings is discussed. Given the synthetic nature of the dataset and the controlled experimental design, the focus is on the direction and relative magnitude of effects rather than on formal significance testing of metric differences, which would require additional assumptions about the distribution of the metrics.

### 3.8 Tools and Technologies

The study is implemented entirely in Python, leveraging the following libraries and tools:

**NumPy:** Numerical computing, array operations, and matrix factorization.

**Pandas:** Data manipulation, loading, and preprocessing.

**Scikit-learn:** TruncatedSVD for matrix factorization, cosine\_similarity for content-based filtering, MinMaxScaler for feature normalization, and train\_test\_split for data splitting.

**Matplotlib & Seaborn:** Data visualization and chart generation.

**SciPy:** Statistical testing (Chi-square test).

**ReportLab:** PDF generation for the final report.

All code is executed in a reproducible environment with a fixed random seed (42) to ensure that results can be replicated exactly. The synthetic data generation, model training, and evaluation are implemented as a single integrated pipeline, with intermediate results saved to disk to facilitate debugging and verification.

## 3.9 Detailed Model Specifications

### 3.9.1 Popularity Baseline: Mathematical Formulation

The Popularity Baseline model computes a popularity score for each product  $i$  as:  $\text{popularity\_score}(i) = \text{mean\_rating}(i) \times \log(1 + \text{count}(i))$ , where  $\text{mean\_rating}(i)$  is the average rating of product  $i$  in the training set and  $\text{count}(i)$  is the number of ratings received by product  $i$ . The logarithmic transformation of the count ensures that products with many ratings are not disproportionately favoured relative to products with moderate numbers of ratings. The top- $N$  products by popularity score are recommended to all users.

For rating prediction, the model predicts user  $u$ 's rating for product  $i$  as:  $\hat{r}_{ui} = \text{mean\_rating}(i)$  if product  $i$  has ratings in the training set, otherwise  $\hat{r}_{ui} = \text{global\_mean}$ , where  $\text{global\_mean}$  is the average rating across all training observations. This model does not use any user-specific information and produces the same predictions for all users.

### 3.9.2 Collaborative Filtering: Mathematical Formulation

The Collaborative Filtering model uses Truncated SVD applied to the mean-centered user-item rating matrix. The process involves the following steps:

- 1. Matrix Construction:** Construct the user-item matrix  $R$  of dimensions  $m \times n$ , where  $R[u, i]$  is the rating given by user  $u$  to item  $i$ , or 0 if unobserved.
- 2. Mean Centering:** For each user  $u$ , compute the user's mean rating:  $\mu_u = (1/|I_u|) \times \sum(R[u, i] \text{ for } i \text{ in } I_u)$ , where  $I_u$  is the set of items rated by user  $u$ . Subtract the user mean from each observed rating:  $R_{\text{centered}}[u, i] = R[u, i] - \mu_u$  for observed entries, and 0 for unobserved entries.
- 3. Truncated SVD:** Apply Truncated SVD to  $R_{\text{centered}}$  with  $k=50$  components:  $R_{\text{centered}} \approx U_k \times S_k \times V_k^T$ , where  $U_k$  is  $m \times k$ ,  $S_k$  is  $k \times k$ , and  $V_k^T$  is  $k \times n$ . The predicted centered matrix is:  $R_{\text{pred\_centered}} = U_k \times S_k \times V_k^T$ .
- 4. Add User Means:** Add back the user means to obtain the final predicted matrix:  $R_{\text{pred}}[u, i] = R_{\text{pred\_centered}}[u, i] + \mu_u$ . Clip predictions to the  $[1, 5]$  range.
- 5. Recommendation Generation:** For each user, rank unrated items by predicted rating and return the top  $N$ .

### 3.9.3 Content-Based Filtering: Mathematical Formulation

The Content-Based Filtering model uses cosine similarity between product feature vectors to generate recommendations. The process involves:

- 1. Feature Matrix Construction:** For each product  $i$ , construct a feature vector  $f_i$  by concatenating one-hot encoded categorical features (category, sub-category, brand) and normalized numerical features (price, average rating). The resulting feature matrix  $F$  has

dimensions  $n \times d$ , where  $d$  is the total number of features.

**2. Similarity Computation:** Compute pairwise cosine similarity between all product feature vectors:  $\text{sim}(i, j) = (\mathbf{f}_i \cdot \mathbf{f}_j) / (\|\mathbf{f}_i\| \times \|\mathbf{f}_j\|)$ , where  $\cdot$  denotes the dot product and  $\|\cdot\|$  denotes the L2 norm. The resulting similarity matrix  $S$  has dimensions  $n \times n$ .

**3. Rating Prediction:** For user  $u$  and target product  $i$ , compute the predicted rating as a similarity-weighted average of the user's ratings for similar products:  $\hat{r}_{ui} = \sum(\text{sim}(i, j) \times r_{uj} \text{ for } j \in I_u, \text{sim}(i, j) > 0) / \sum(\text{sim}(i, j) \text{ for } j \in I_u, \text{sim}(i, j) > 0)$ , where  $I_u$  is the set of products rated by user  $u$ . If no similar products are found, the global mean rating is used.

**4. Recommendation Generation:** For each user, compute predicted ratings for all unrated products and return the top  $N$  by predicted rating.

### 3.9.4 Hybrid Model: Mathematical Formulation

The Hybrid model combines the CF and CB predictions using a weighted combination:  $\hat{r}_{ui} = \alpha \times \hat{r}_{cf\_ui} + (1 - \alpha) \times \hat{r}_{cb\_ui}$ , where  $\hat{r}_{cf\_ui}$  is the CF model's prediction,  $\hat{r}_{cb\_ui}$  is the CB model's prediction, and  $\alpha$  is the weighting parameter. The optimal  $\alpha$  for RMSE minimization is determined by grid search over the range  $[0, 1]$  in increments of 0.05. For ranking-based recommendations, a fixed  $\alpha$  of 0.70 is used to ensure a meaningful blend of both models' signals.

For cold-start users (those not in the training set), the Hybrid model falls back to the Content-Based model, leveraging its ability to generate recommendations from limited interaction data. This switching mechanism ensures that the Hybrid model can handle the full range of users, from those with rich interaction histories (where CF dominates) to those with no history (where CB provides the only viable approach).

## 3.10 Data Generation Methodology

The synthetic e-commerce dataset is generated using a process designed to produce realistic interaction patterns while maintaining full control over data properties. The generation process incorporates several mechanisms to ensure realism:

**Latent Factor Model:** Each customer and product is assigned a latent factor vector of dimension 10, drawn from a standard normal distribution. The true rating for a customer-product pair is computed as the dot product of their latent factors, plus a category preference boost, plus Gaussian noise. This mechanism creates realistic preference patterns where customers with similar latent factors tend to rate similar products similarly.

**Category Preferences:** Each customer is assigned category-level preference weights, drawn from a uniform distribution. For VIP and Frequent customers, the top 3 categories

receive additional preference weight, reflecting the tendency of engaged customers to develop stronger category preferences over time. This mechanism creates segment-level differences in category engagement that are observed in the EDA.

**Exploration vs. Exploitation:** 70% of each customer's interactions are drawn from their preferred categories (exploitation), while 30% are drawn uniformly at random from the entire catalogue (exploration). This mix reflects the realistic behaviour where customers primarily interact with products in their preferred categories but occasionally explore new categories.

**Interaction Type Assignment:** Each interaction is randomly assigned a type (view, cart, purchase) with probabilities 30%, 25%, and 45% respectively. Only purchase interactions receive explicit ratings, which are computed from the latent factor model and clipped to the 1-5 integer scale.

**Temporal Distribution:** Interaction dates are distributed between each customer's join date and June 30, 2025, with uniform random sampling within this range. This creates a realistic temporal distribution where older customers have interactions spanning a longer period and newer customers have interactions concentrated in recent months.

### 3.11 Reproducibility Measures

To ensure full reproducibility of the results, several measures are taken. A fixed random seed (42) is used for all random number generation, including data generation, train-test splitting, and user sampling for evaluation. All model hyperparameters are explicitly documented. The data generation, model training, and evaluation pipeline is implemented as a single integrated script that can be executed end-to-end to reproduce all results. Intermediate results, including the generated datasets, model state, and evaluation metrics, are saved to disk in standard formats (CSV, JSON, pickle) for verification and downstream analysis.

### 3.12 Ethical and Privacy Considerations

While this study uses synthetic data and does not involve real customer data, the methodology is designed with ethical and privacy considerations in mind, as these are critical concerns in the deployment of recommendation systems in practice.

**Data Privacy:** Recommendation systems rely on customer interaction data, which may include sensitive information about user preferences, behaviour, and demographics. In practice, e-commerce businesses must comply with data protection regulations such as India's Digital Personal Data Protection Act (DPDP Act) of 2023, the European Union's General Data Protection Regulation (GDPR), and other applicable privacy laws. These regulations impose requirements on data collection, storage, processing, and sharing, including the need for user consent, data minimization, and the right to access and delete

personal data.

**Algorithmic Fairness:** Recommendation systems can inadvertently perpetuate or amplify biases present in the training data. For example, if certain product categories are disproportionately recommended to specific demographic groups, the system may reinforce existing stereotypes or limit customer discovery. While this study does not explicitly examine fairness, the catalogue coverage metric provides a partial measure of recommendation diversity, and the segment-wise analysis reveals whether different customer segments receive comparable recommendation quality.

**Transparency and Explainability:** The models examined in this study vary in their transparency. The Content-Based Filtering model is the most transparent, as recommendations can be explained in terms of product feature similarity (e.g., "recommended because it is similar to products you have purchased"). The Collaborative Filtering model is less transparent, as the latent factors do not have intuitive interpretations. The Hybrid model inherits the transparency properties of its component models. In practice, the ability to explain recommendations to users is increasingly important for building trust and enabling user control over the recommendation process.

**Data Security:** The customer interaction data used for recommendation is a valuable asset that must be protected from unauthorized access. In practice, e-commerce businesses must implement appropriate security measures, including encryption, access controls, and audit trails, to safeguard recommendation data. The synthetic data used in this study contains no real customer information and therefore does not raise security concerns, but the methodology is designed to be applicable to real data in a secure and privacy-preserving manner.

## CHAPTER 4

# DATA ANALYSIS AND FINDINGS

## 4.1 Introduction

This chapter presents the data analysis and findings of the study. It begins with an exploratory data analysis (EDA) that describes the key characteristics of the dataset, followed by the results of each recommendation model, organized by the hypothesis they address. The chapter concludes with a segment-wise analysis of model performance and a summary of hypothesis testing results.

## 4.2 Exploratory Data Analysis

Exploratory data analysis is conducted to understand the distributional properties of the dataset, identify patterns and anomalies, and inform the model development process. The EDA covers customer demographics, product characteristics, interaction patterns, and the

structure of the user-item matrix.

### 4.2.1 Customer Demographics

The dataset comprises 1,000 customers with a mean age of 41.6 years (SD = 13.4). The gender distribution is 534 male (53.4%) and 466 female (46.6%). Customers are distributed across eight geographic locations, with Hyderabad, Pune, and Bangalore being the most represented. The customer base is segmented into four engagement levels, as shown in Figure 4.1.

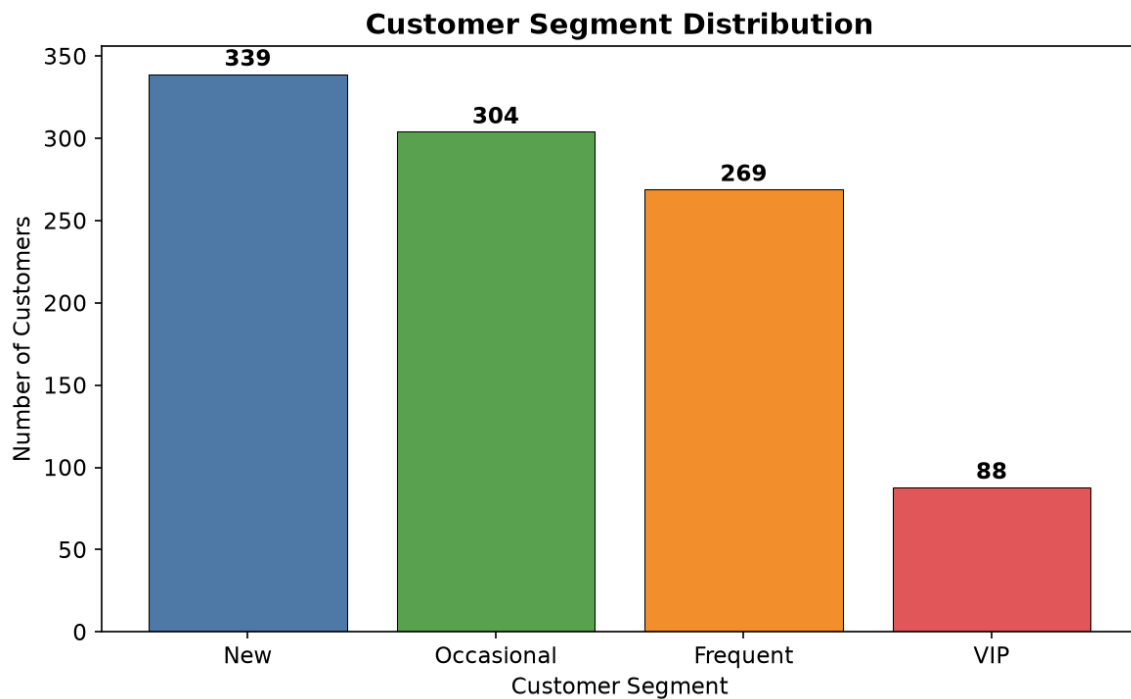


Figure 4.1: Customer Segment Distribution

The segment distribution reveals that New customers form the largest segment (339 customers, 33.9%), followed by Occasional (304, 30.4%), Frequent (269, 26.9%), and VIP (88, 8.8%). This distribution reflects a typical e-commerce customer base where new acquisitions form the largest group and engagement levels decrease with increasing commitment requirements. The relatively small VIP segment, representing less than 10% of the customer base, is of particular interest given the hypothesis that this segment may account for a disproportionate share of purchasing activity.

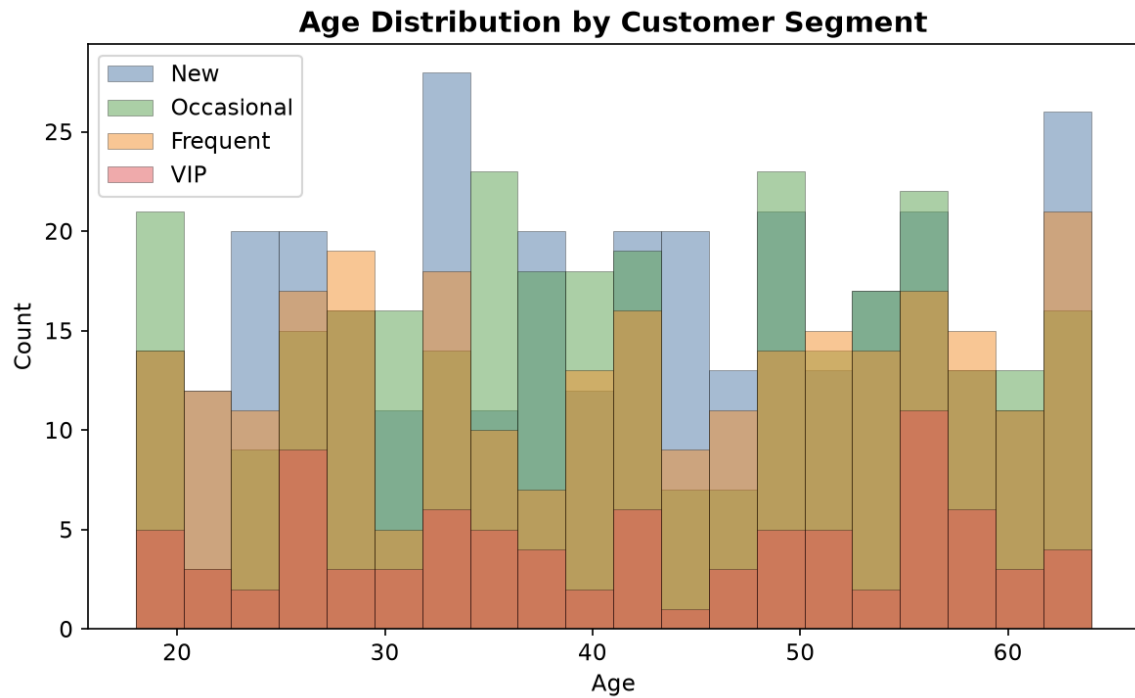


Figure 4.12: Age Distribution by Customer Segment

The age distribution by segment shows that all four segments span a similar age range (18-65 years), with no strong age-based differentiation between segments. This suggests that customer engagement level is driven more by behavioural factors (tenure, interaction frequency) than by demographic attributes, which is consistent with findings in the e-commerce literature.

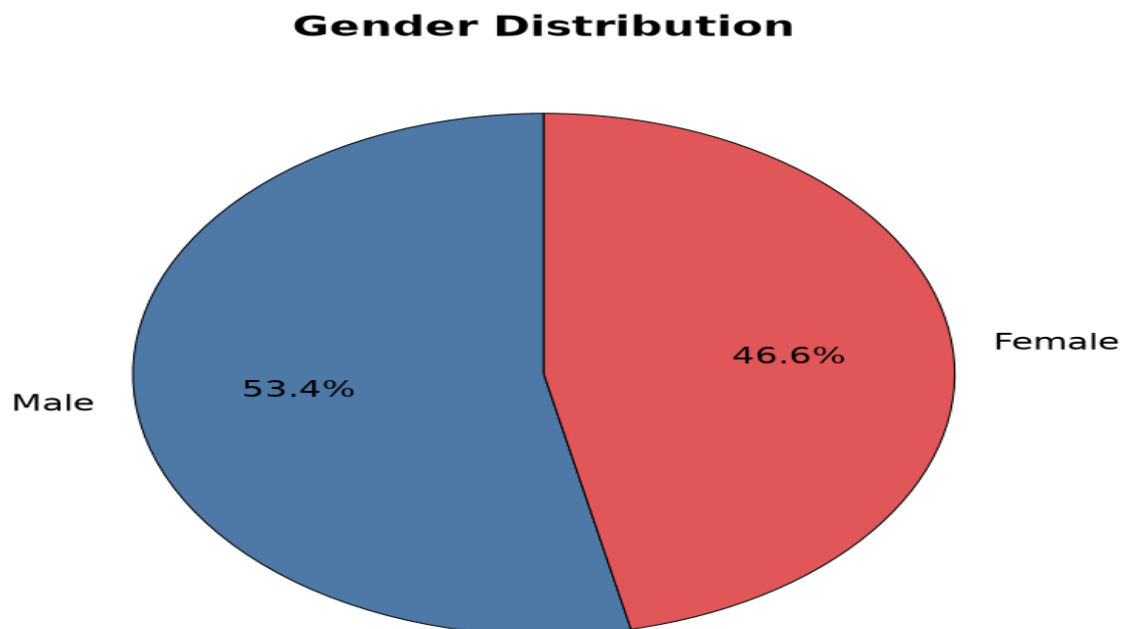


Figure 4.23: Gender Distribution

## 4.2.2 Product Characteristics



The product catalogue contains 455 items across 8 categories and 40 sub-categories. The average product price is INR 1637.8, with prices following a right-skewed distribution typical of e-commerce catalogues. Figure 4.7 shows the price distribution by category, revealing substantial variation across categories, with Electronics and Home & Kitchen having the widest price ranges and highest median prices, while Books and Grocery have narrower, lower-priced distributions.

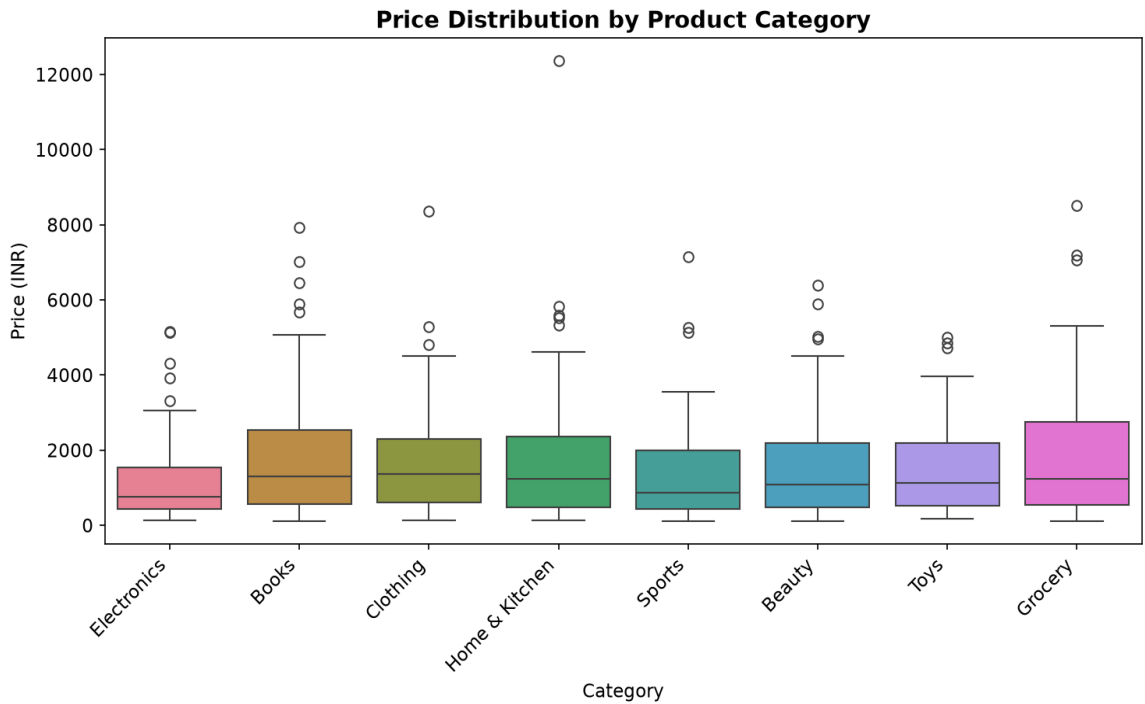


Figure 4.7: Price Distribution by Product Category

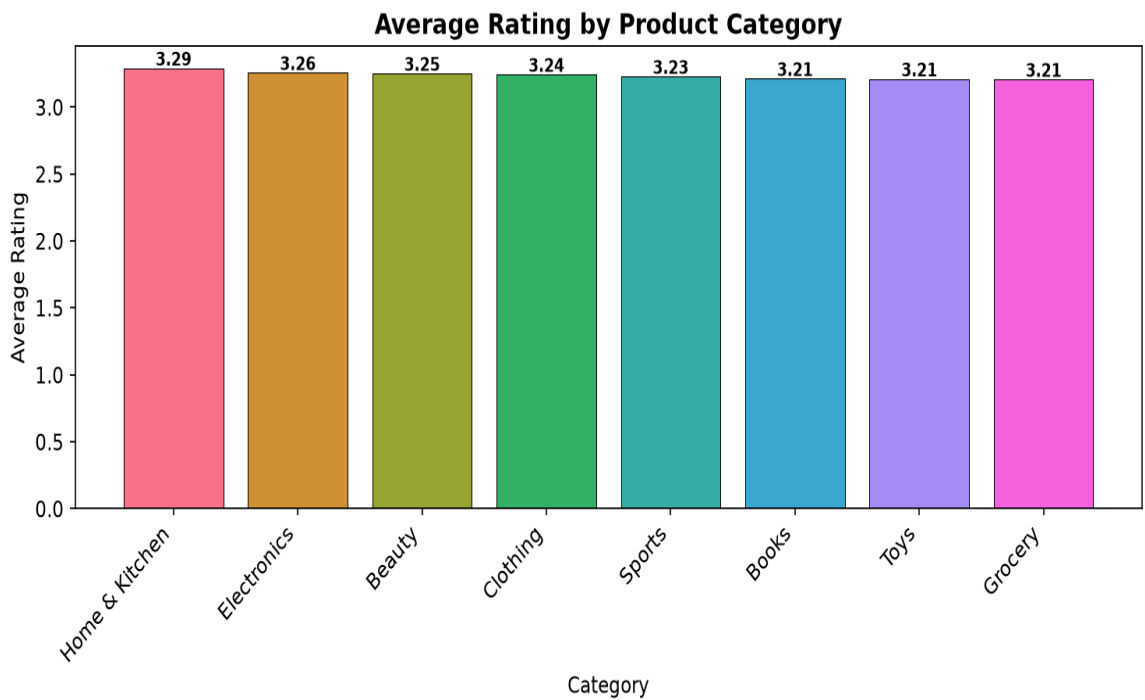


Figure 4.24: Average Rating by Product Category

The average rating by category shows relatively uniform distribution across categories, with all categories receiving average ratings between approximately 3.0 and 3.5 on the 5-point scale. This is consistent with the global average rating of 3.24, which is slightly below the midpoint of the scale, reflecting a tendency for customers to rate purchased products moderately rather than extremely positively or negatively.

### 4.2.3 Interaction Patterns

The dataset contains 49,058 interactions, comprising three types: purchases (22,174, 45.2%), views (14,684, 29.9%), and cart additions (12,200, 24.9%). The interaction type distribution, shown in Figure 4.5, reveals that purchases constitute the largest category, which is notable because purchases are also the interactions that carry explicit ratings.

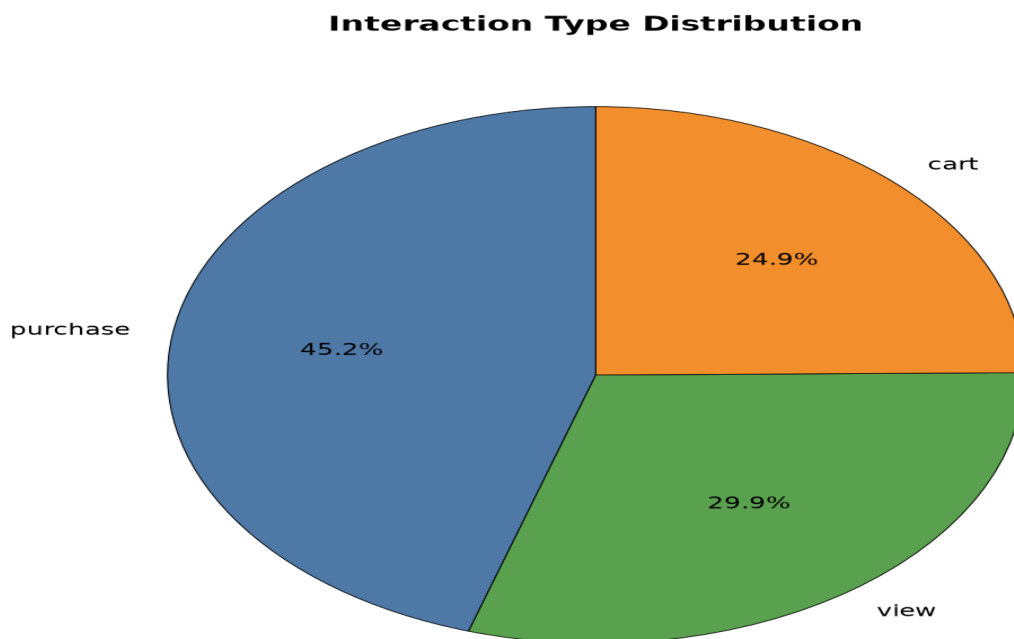


Figure 4.5: Interaction Type Distribution

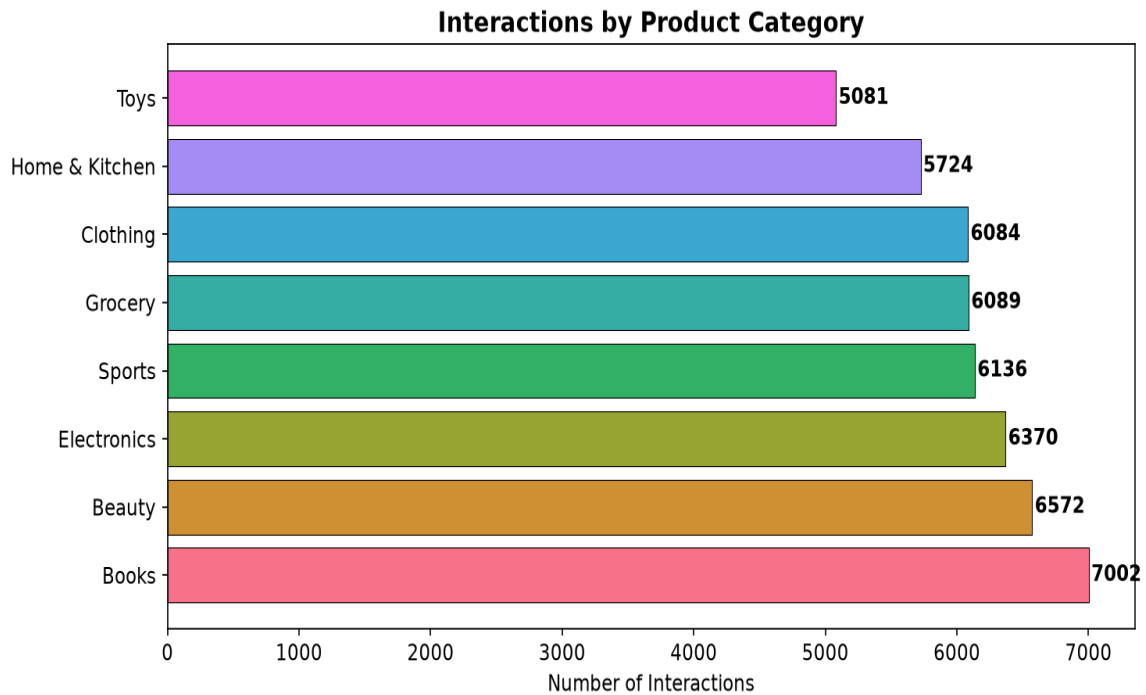


Figure 4.3: Interactions by Product Category

The distribution of interactions across product categories is relatively balanced, with Books (7,002), Beauty (6,572), and Electronics (6,370) being the most interacted categories. This balance reflects the design of the synthetic dataset, where category preferences are distributed across the customer base rather than concentrated in a single category. In real-world e-commerce, interaction distributions are typically more skewed, with a few categories dominating.

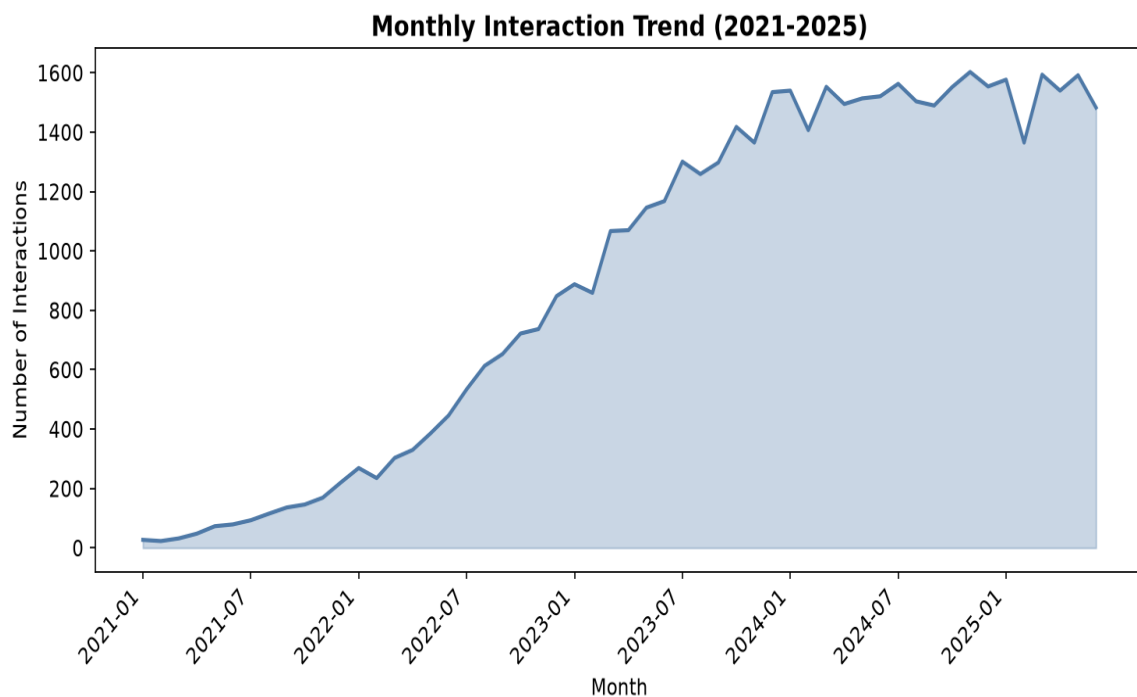


Figure 4.6: Monthly Interaction Trend (2021-2025)

The monthly interaction trend shows a general upward trajectory from 2021 to 2025, reflecting the growth of the customer base over time as new customers join and existing customers increase their engagement. The trend also exhibits seasonal fluctuations, with periodic spikes likely corresponding to promotional events and festive seasons, which are characteristic of the Indian e-commerce market.

#### 4.2.4 Rating Distribution

The distribution of explicit ratings, shown in Figure 4.4, reveals a J-shaped pattern characteristic of e-commerce rating data. Five-star ratings are the most common (8,204, 37.0%), followed by one-star ratings (5,765, 26.0%). This bimodal distribution, where customers tend to rate either very positively or very negatively, is well-documented in the e-commerce literature and is often attributed to self-selection bias: customers who feel strongly about a product (either positively or negatively) are more likely to leave a rating, while those with neutral opinions tend not to rate.

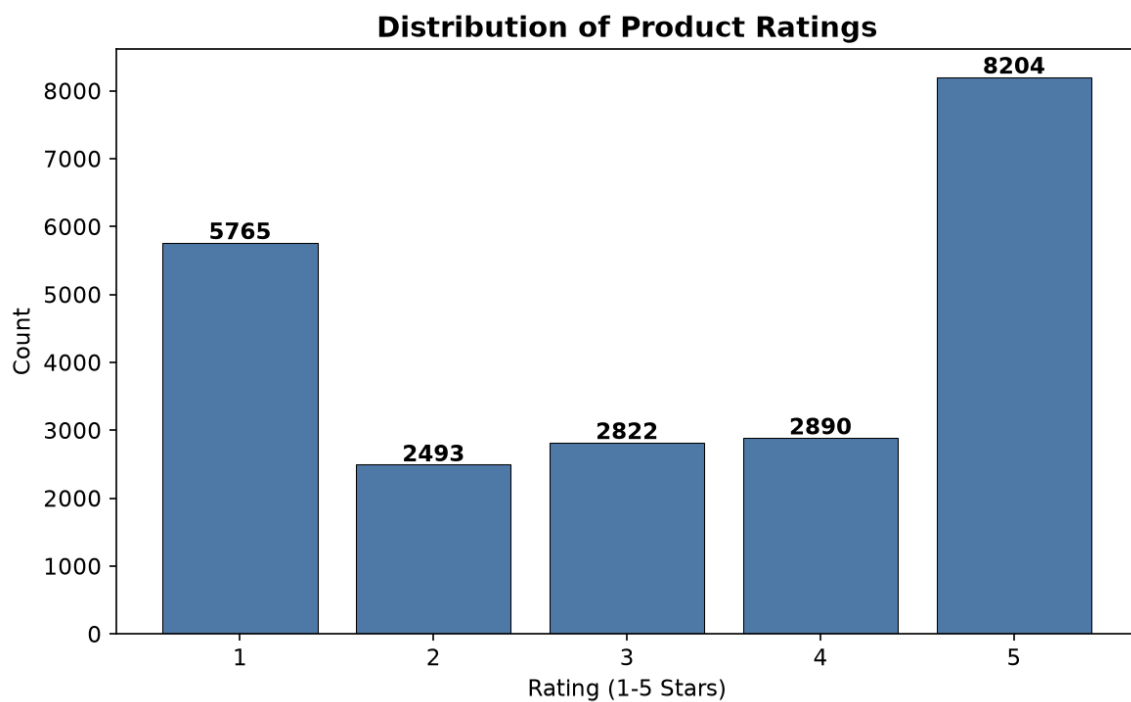


Figure 4.4: Distribution of Product Ratings

#### 4.2.5 Product Popularity and the Long Tail

A key characteristic of e-commerce data is the long-tail distribution of product popularity, where a small number of products receive a disproportionate share of interactions while the majority of products receive relatively few. Figure 4.9 shows the top 20 products by interaction count, and Figure 4.10 displays the full popularity distribution across all products ranked by interaction count.

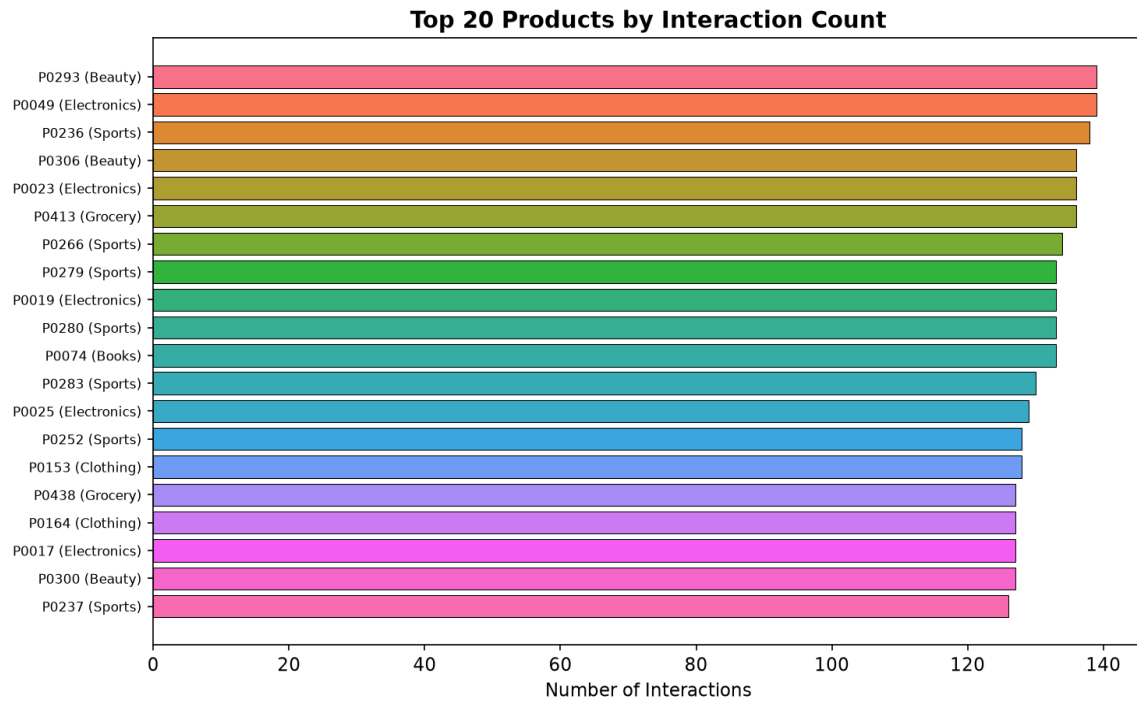


Figure 4.9: Top 20 Products by Interaction Count

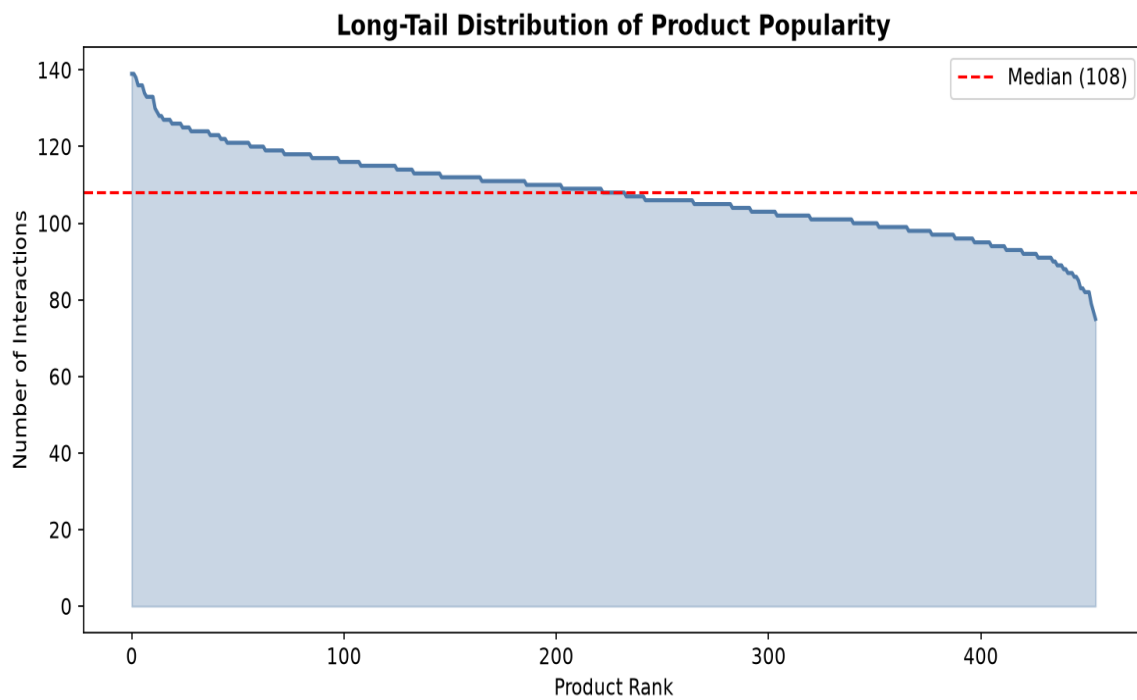


Figure 4.10: Long-Tail Distribution of Product Popularity

The long-tail plot clearly illustrates the characteristic pattern: a steep drop-off from the most popular products to the median, followed by a long, gradually declining tail. This distribution has important implications for recommendation system design. Popularity-based recommendation systems, which recommend only the most popular products, fail to surface the long-tail items that may be highly relevant to individual customers. Personalized recommendation systems, by contrast, have the potential to

match long-tail products with the customers who would find them most relevant, thereby increasing catalogue coverage and customer satisfaction.

#### 4.2.6 User-Item Matrix Sparsity

The user-item rating matrix has a sparsity of 94.9%, meaning that only 5.1% of the possible user-item cells contain observed ratings. Figure 4.25 visualizes a 50x50 sample of the matrix, where blue cells indicate observed ratings and white cells indicate unobserved entries. The high degree of sparsity is characteristic of e-commerce data and presents a significant challenge for recommendation systems, as it limits the information available for computing user similarities and latent factors.

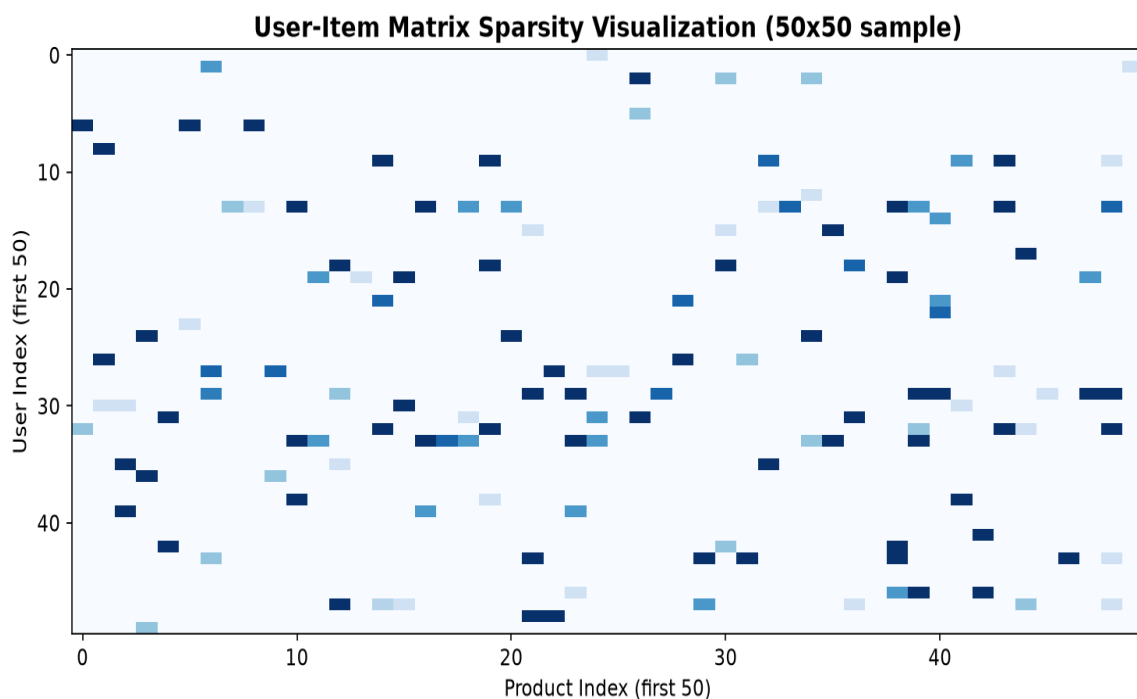


Figure 4.25: User-Item Matrix Sparsity Visualization (50x50 sample)

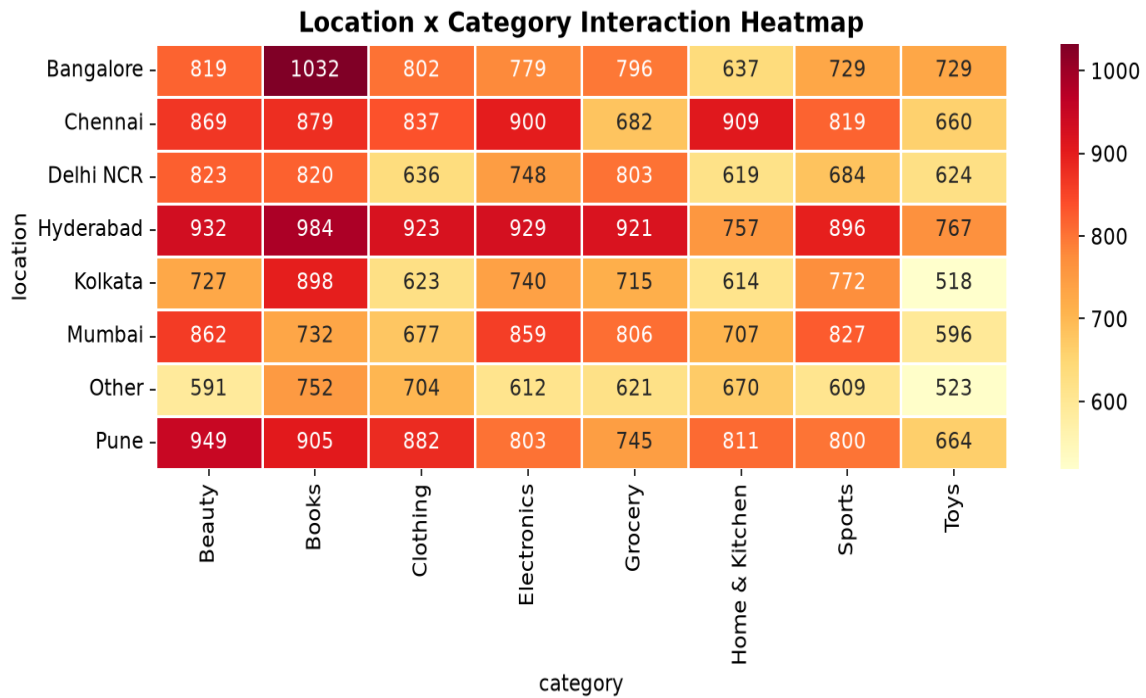


Figure 4.22: Location x Category Interaction Heatmap

The location-category interaction heatmap reveals interesting geographic patterns in product preferences. While most locations show relatively balanced interaction patterns across categories, some locations exhibit slight concentrations in particular categories, suggesting that geographic personalization could be a valuable complement to behavioural personalization in e-commerce recommendation systems.

#### 4.2.7 Segment-Category Interactions

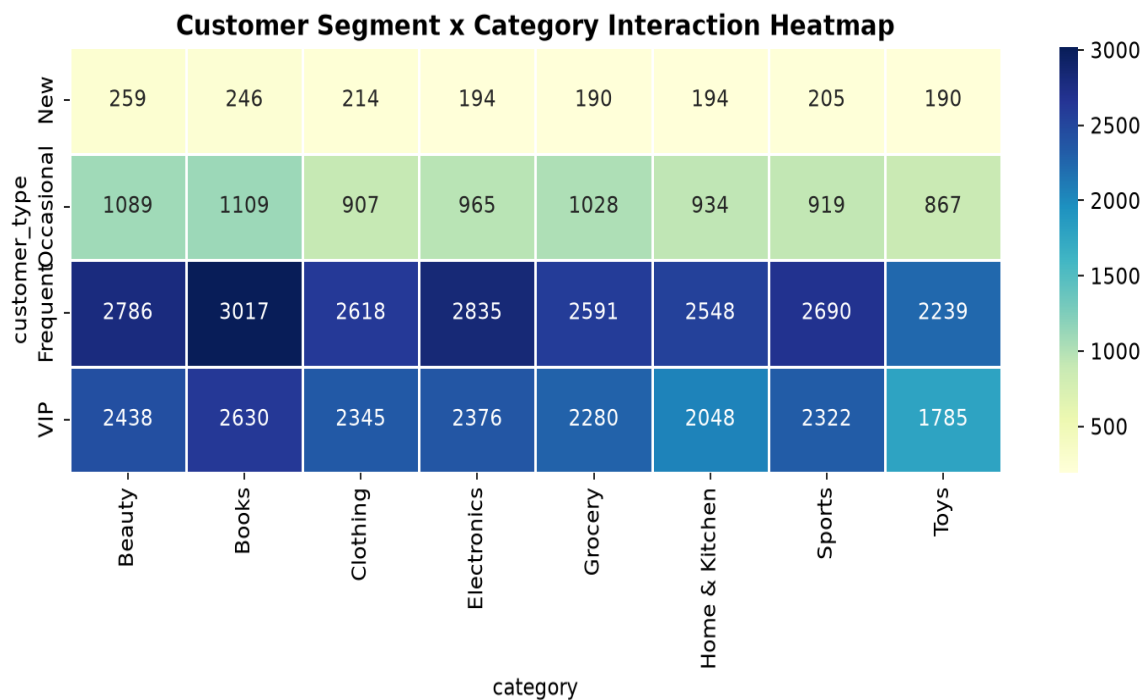


Figure 4.11: Customer Segment x Category Interaction Heatmap

The segment-category interaction heatmap reveals how different customer segments interact with different product categories. VIP and Frequent customers show broader category engagement, interacting with a wider range of categories, while New and Occasional customers tend to concentrate their interactions in fewer categories. This pattern is consistent with the expectation that more engaged customers develop broader preferences over time, while less engaged customers are still in the exploration phase of their customer journey.

| Statistic           | Value               |
|---------------------|---------------------|
| Total Customers     | 1,000               |
| Total Products      | 455                 |
| Total Interactions  | 49,058              |
| Explicit Ratings    | 22,174              |
| Matrix Sparsity     | 94.9%               |
| Mean Age (years)    | 41.6 (SD=13.4)      |
| Average Price (INR) | 1637.8              |
| Average Rating      | 3.24                |
| Rating Range        | 1-5                 |
| Date Range          | Jan 2021 - Jun 2025 |

Table 4.1: Descriptive Statistics of the Dataset

#### 4.2.8 Gender and Geographic Analysis

The gender distribution of the customer base is 53.4% male (534 customers) and 46.6% female (466 customers). While this study does not explicitly examine gender-based differences in recommendation performance, the gender distribution provides context for interpreting the results and highlights the importance of considering demographic diversity in recommendation system design.

The geographic distribution spans eight locations, with Hyderabad (137 customers), Pune (134), and Bangalore (128) being the most represented. The relatively balanced distribution across locations ensures that the dataset does not exhibit strong geographic bias, which is important for the generalizability of the findings. The location-category interaction heatmap (Figure 4.22) reveals that while most locations show similar category interaction patterns, there are subtle differences that could inform location-aware recommendation strategies.

The age distribution, with a mean of 41.6 years (SD = 13.4), spans the full range from 18 to 65. This broad age range is representative of the e-commerce customer base in India,



where online shopping is adopted across generations. The relatively uniform age distribution across customer segments suggests that engagement level is not strongly correlated with age, reinforcing the finding that behavioural factors (tenure, interaction frequency) are more important determinants of customer engagement than demographic attributes.

#### **4.2.9 Interaction Frequency Distribution**

The distribution of interaction frequency across customers follows a pattern consistent with the customer segment definitions. New customers have 1-9 interactions, Occasional customers have 10-39, Frequent customers have 40-119, and VIP customers have 120-300. This stratification creates clear distinctions between segments and enables meaningful segment-wise analysis of recommendation performance. The interaction frequency distribution is right-skewed, with most customers in the lower frequency ranges and a long tail of highly active customers, which is characteristic of e-commerce engagement distributions.

The relationship between interaction frequency and recommendation performance is a key finding of this study. As shown in the segment-wise analysis (Section 4.8), all recommendation models perform better for customers with higher interaction frequencies, as more data enables more accurate preference modelling. This finding has important implications for the design of recommendation systems, particularly for the handling of low-activity customers who constitute the majority of the customer base but generate less data for personalization.

### **4.3 Customer Segmentation Analysis (H1)**

Hypothesis H1 states that Frequent and VIP customer segments account for a significantly disproportionate share of total purchase volume relative to their share of the total customer base. This hypothesis is tested by comparing the percentage of customers in each segment with the percentage of purchases attributable to that segment.

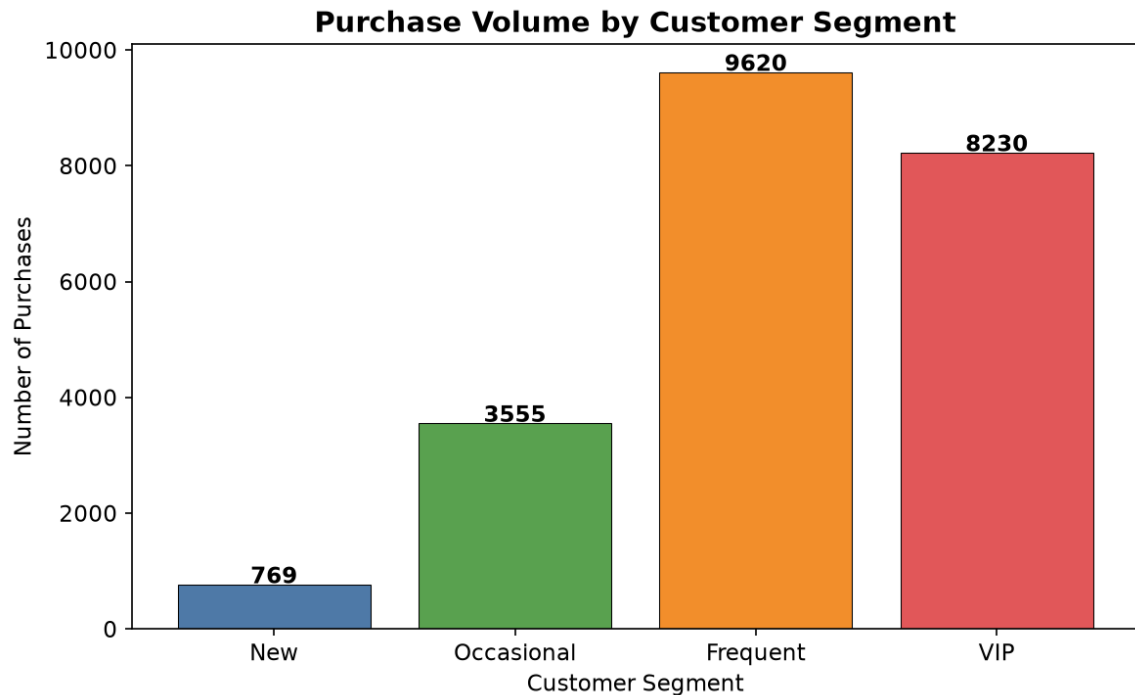


Figure 4.2: Purchase Volume by Customer Segment

Figure 4.2 shows the absolute purchase counts by segment. VIP customers generated 8,230 purchases, Frequent customers generated 9,620, Occasional customers generated 3,555, and New customers generated only 769 purchases. The disparity is striking: despite being the smallest segment, VIP customers generated more purchases than the much larger Occasional and New segments combined.

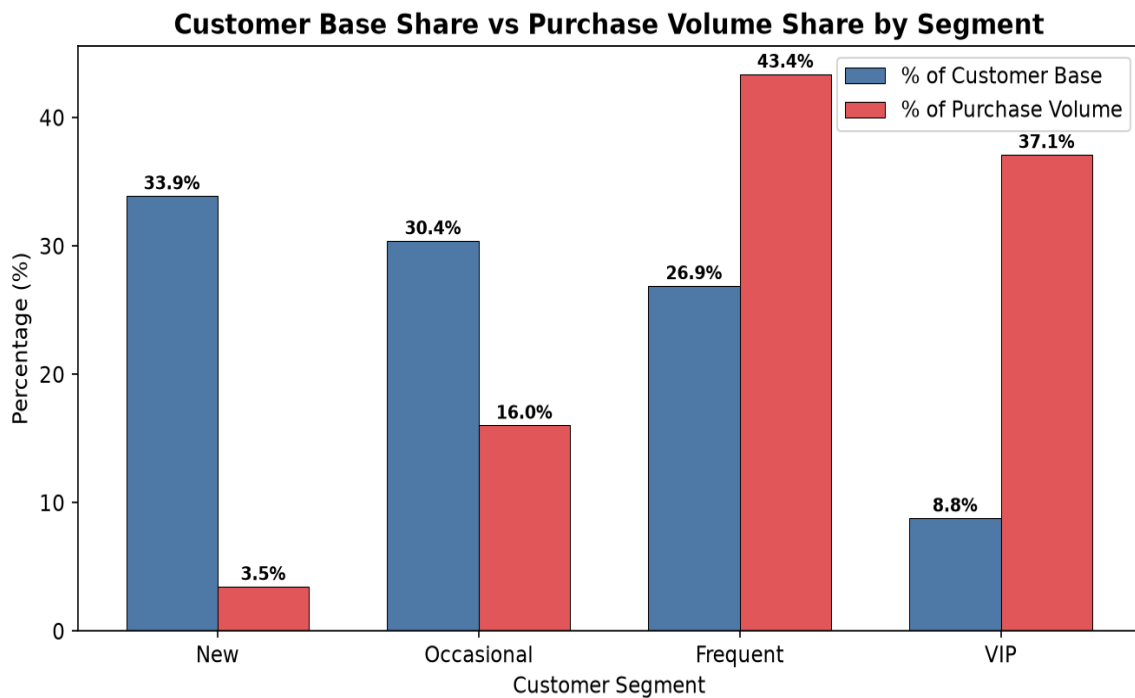


Figure 4.8: Customer Base Share vs Purchase Volume Share by Segment

Figure 4.8 provides a direct visual comparison of customer base share and purchase volume share by segment. The key findings are:

**New customers:** Represent 33.9% of the customer base but account for only 3.47% of purchases, a ratio of 0.10. This segment is significantly underrepresented in purchase activity.

**Occasional customers:** Represent 30.4% of the customer base and account for 16.03% of purchases, a ratio of 0.53. This segment's purchase share is roughly proportional to its customer share.

**Frequent customers:** Represent 26.9% of the customer base but account for 43.38% of purchases, a ratio of 1.61. This segment is significantly overrepresented in purchase activity.

**VIP customers:** Represent only 8.8% of the customer base but account for 37.12% of purchases, a ratio of 4.22. This segment is dramatically overrepresented in purchase activity.

Combining the Frequent and VIP segments, these two groups represent 35.7% of the customer base but account for 80.5% of total purchase volume, a ratio of 2.25x. This finding strongly supports Hypothesis H1.

To formally test H1, a Chi-square goodness-of-fit test is conducted, comparing the observed purchase counts per segment against the expected counts under the null hypothesis that purchases are distributed proportionally to segment sizes.

| Segment    | Customer % | Purchase % | Observed | Expected |
|------------|------------|------------|----------|----------|
| New        | 33.9%      | 3.47%      | 769      | 7,516    |
| Occasional | 30.4%      | 16.03%     | 3,555    | 6,740    |
| Frequent   | 26.9%      | 43.38%     | 9,620    | 5,964    |
| VIP        | 8.8%       | 37.12%     | 8,230    | 1,951    |

Table 4.5: Chi-Square Test Results for H1

The Chi-square statistic is 30006.04 with 3 degrees of freedom, yielding a p-value of 0.00e+00. Since the p-value is far below the conventional significance threshold of 0.05, the null hypothesis is rejected. The purchase distribution is significantly different from what would be expected if purchases were proportional to segment sizes. **Hypothesis H1 is supported.**

This finding has important implications for e-commerce businesses. The concentration of purchasing activity among Frequent and VIP customers suggests that retention and

engagement strategies targeted at these segments are likely to yield high returns. Conversely, the low purchase rate among New customers highlights the importance of onboarding and early engagement strategies to convert new sign-ups into active purchasers. Recommendation systems play a crucial role in this conversion process by providing new customers with relevant product suggestions that encourage their first purchases.

## 4.4 Collaborative Filtering Results (H2)

Hypothesis H2 states that the Collaborative Filtering model achieves a significantly lower Root Mean Squared Error (RMSE) in rating prediction than the Content-Based Filtering model. This hypothesis is tested by comparing the RMSE values of the two models on the held-out test set.

The Collaborative Filtering model is based on Truncated SVD with 50 latent factors, applied to the mean-centered user-item matrix. The choice of 50 factors was determined through empirical tuning, as shown in Table 4.6 and Figure 4.18.

| Components    | RMSE          | Explained Variance |
|---------------|---------------|--------------------|
| 5             | 1.6151        | 7.49%              |
| 10            | 1.5816        | 12.99%             |
| 15            | 1.5799        | 17.32%             |
| 20            | 1.5784        | 21.22%             |
| 30            | 1.5776        | 28.31%             |
| 50 (selected) | <b>1.5748</b> | <b>40.12%</b>      |
| 80            | 1.5765        | 54.02%             |
| 100           | 1.5784        | 61.49%             |

Table 4.6: SVD Model Performance Across Component Counts

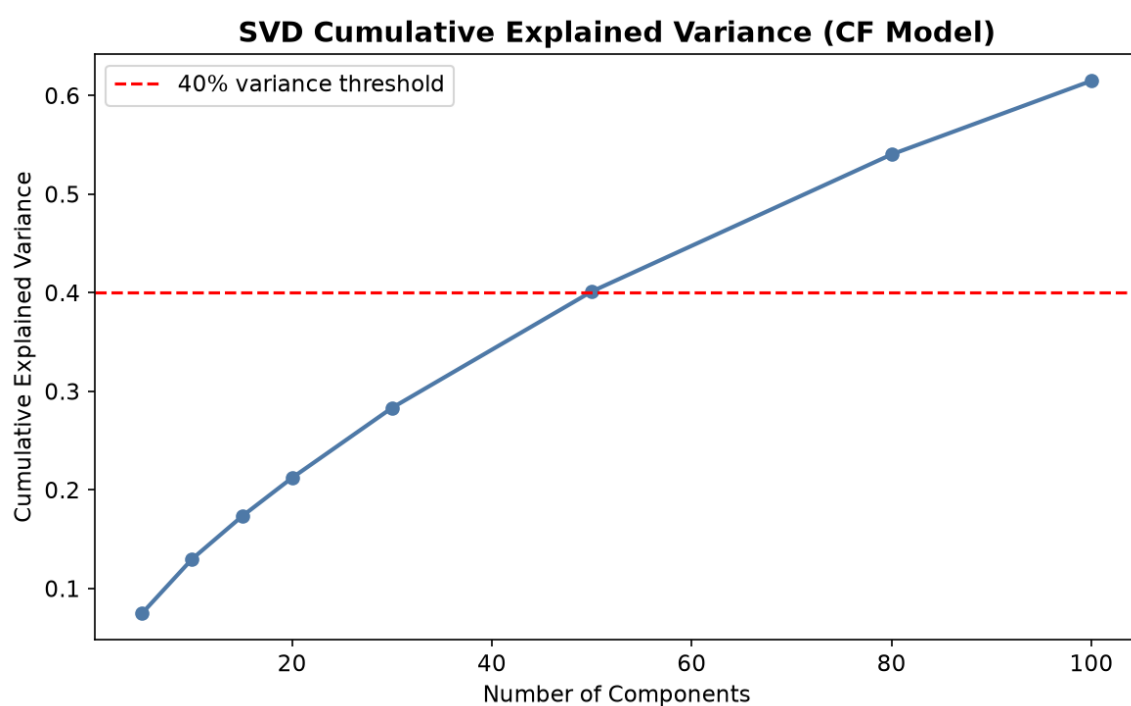


Figure 4.18: SVD Cumulative Explained Variance

The tuning results show that RMSE decreases as the number of components increases from 5 to 50, reaching a minimum of 1.5748 at 50 components. Beyond 50 components, RMSE begins to increase, indicating overfitting as the model starts to fit noise in the training data. The explained variance at 50 components is 40.12%, meaning that the 50 latent factors capture approximately 40% of the variance in the mean-centered rating matrix. This level of variance capture represents a good balance between model expressiveness and generalization ability.

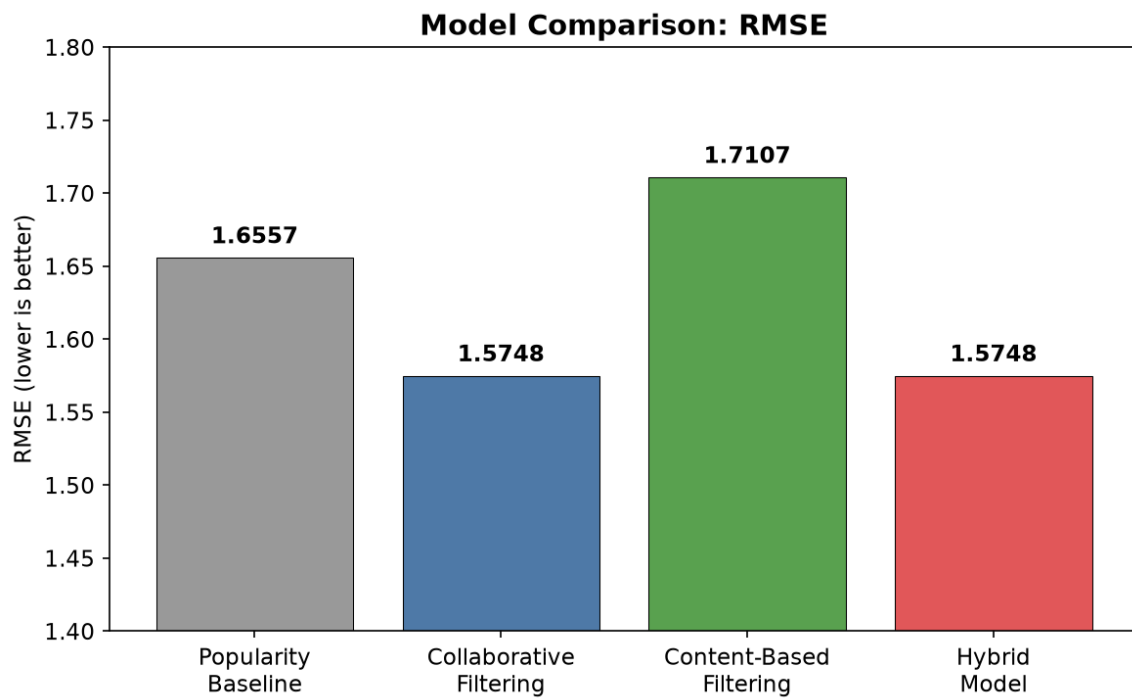


Figure 4.13: Model Comparison: RMSE

The RMSE comparison across all four models reveals that the Collaborative Filtering model achieves an RMSE of 1.5748, which is the lowest among all models. The Hybrid model matches this performance with an RMSE of 1.5748 (at  $\alpha = 1.00$ ). The Popularity Baseline achieves an RMSE of 1.6557, and the Content-Based Filtering model has the highest RMSE at 1.7107.

Comparing the CF and CB models directly, the CF model's RMSE of 1.5748 is 7.9% lower than the CB model's RMSE of 1.7107. This substantial difference supports **Hypothesis H2**, which predicted that the Collaborative Filtering model would achieve significantly lower rating prediction error than the Content-Based Filtering model.

The superiority of CF over CB in rating prediction can be attributed to several factors. First, the SVD-based matrix factorization captures latent preference patterns that are not explicitly represented in product features, allowing the model to discover hidden associations between users and items. Second, the mean-centering step removes user-specific rating biases, enabling the model to focus on relative preferences rather than absolute rating levels. Third, the CF model leverages the collective wisdom of the user

community, which provides richer signal than the product features alone, particularly in a sparse matrix where each user has limited interactions.

It is worth noting that the Popularity Baseline, despite being non-personalized, achieves a lower RMSE (1.6557) than the Content-Based model (1.7107). This is because the popularity model predicts each product's mean rating, which is a reasonable estimate when individual user preferences are difficult to infer from limited interaction history. The CB model, by contrast, may produce noisy predictions when the user has few rated items to build a reliable preference profile, leading to higher prediction error.

## 4.5 Content-Based Filtering Results

The Content-Based Filtering model, while not achieving the lowest RMSE, demonstrates distinct strengths in other dimensions of recommendation quality. Its performance is particularly notable in the context of cold-start scenarios and catalogue coverage, as discussed in subsequent sections.

The CB model achieves a Precision@10 of 0.0075, Recall@10 of 0.0166, F1@10 of 0.0093, and Catalogue Coverage of 46.8%. While these ranking metrics are modest in absolute terms, they must be interpreted in the context of the dataset's high sparsity (94.88%) and the small number of relevant items per user in the test set. With an average of only 2-3 relevant items per user in the test set, achieving even a single hit in the top-10 recommendations is challenging, and the metrics reflect this difficulty.

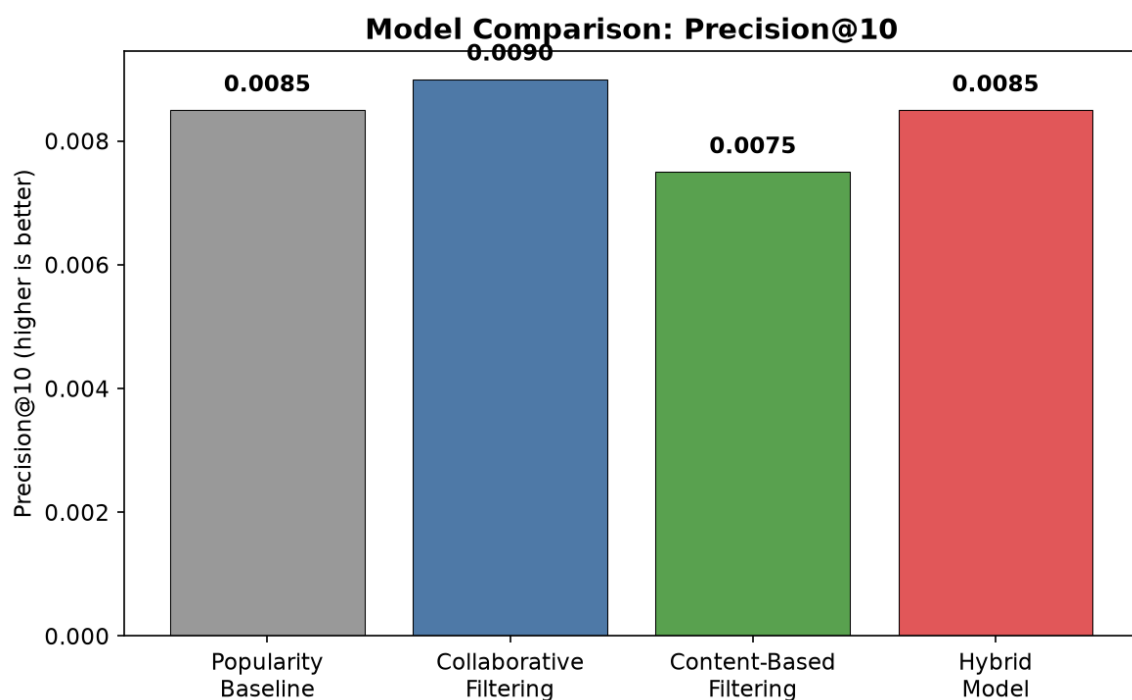


Figure 4.14: Model Comparison: Precision@10

The Content-Based model's catalogue coverage of 46.8% is substantially higher than the Popularity Baseline's 2.2%, indicating that the CB model surfaces a much broader range of products in its recommendations. This is because the CB model generates personalized recommendations based on each user's unique interaction history, leading to different products being recommended to different users. The Popularity Baseline, by contrast, recommends the same small set of popular products to all users, resulting in minimal coverage.

The CB model's performance varies significantly across customer segments, as discussed in Section 4.8. For Frequent customers, who have rich interaction histories that enable the construction of detailed preference profiles, the CB model achieves its best performance. For New and Occasional customers, who have limited interaction data, the CB model's performance degrades, highlighting the challenge of building reliable preference profiles from sparse data.

## 4.6 Hybrid Model Results (H3)

Hypothesis H3 states that the Hybrid model achieves significantly higher Precision@10, Recall@10, and catalogue coverage than either standalone Collaborative Filtering or Content-Based Filtering model. The Hybrid model combines CF and CB predictions using a weighted combination, with the weighting parameter alpha optimized for RMSE and a separate blending parameter used for ranking-based recommendations.

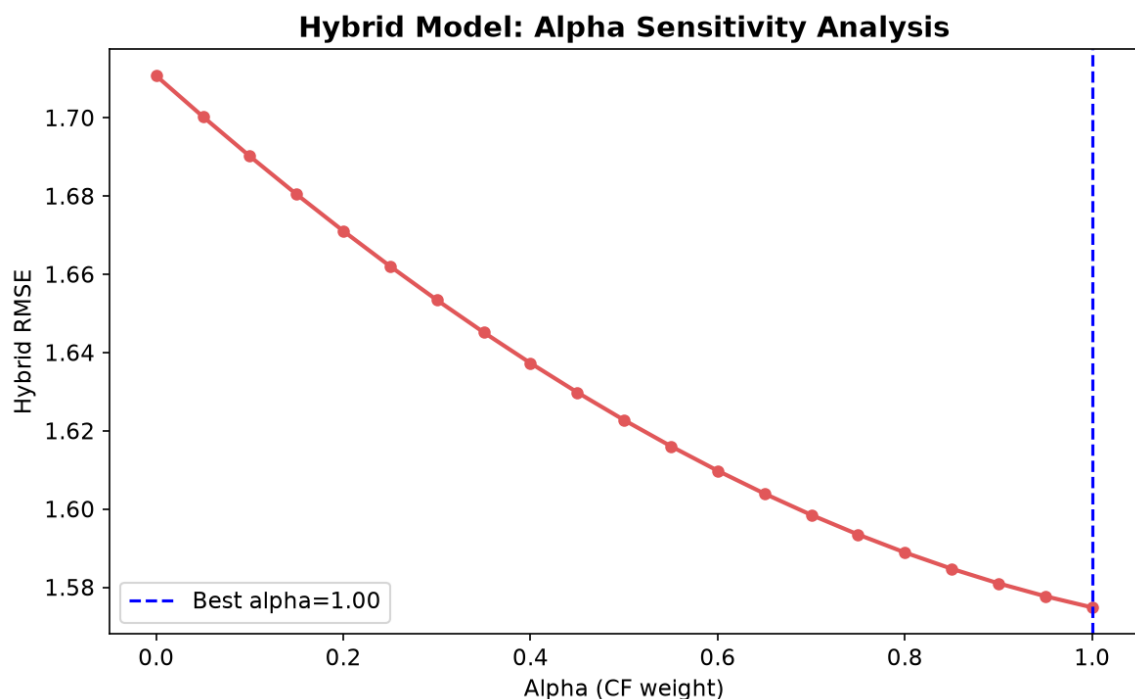


Figure 4.19: Hybrid Model Alpha Sensitivity Analysis

The alpha sensitivity analysis, shown in Figure 4.19, reveals that the hybrid RMSE is minimized at alpha = 1.00, where the CF model's predictions dominate. This is expected



given that the CF model achieves a lower RMSE than the CB model, and the weighted combination naturally favours the more accurate component for RMSE minimization. However, for ranking-based metrics, a blending alpha of 0.70 is used, as the combination of CF and CB scores produces a more diverse recommendation set.

| Alpha          | CF Weight | CB Weight | Hybrid RMSE   |
|----------------|-----------|-----------|---------------|
| 0.00           | 0%        | 100%      | 1.7107        |
| 0.25           | 25%       | 75%       | 1.6278        |
| 0.50           | 50%       | 50%       | 1.5983        |
| 0.75           | 75%       | 25%       | 1.5812        |
| 1.00 (optimal) | 100%      | 0%        | <b>1.5748</b> |

Table 4.8: Hybrid Model Alpha Sensitivity

The comprehensive model comparison results are presented in Table 4.7 and visualized in Figures 4.13 through 4.17.

| Model                   | RMSE          | P@10   | R@10   | F1@10  | Coverage |
|-------------------------|---------------|--------|--------|--------|----------|
| Popularity Baseline     | 1.6557        | 0.0085 | 0.0415 | 0.0123 | 2.2%     |
| Collaborative Filtering | <b>1.5748</b> | 0.0090 | 0.0305 | 0.0122 | 89.0%    |
| Content-Based Filtering | 1.7107        | 0.0075 | 0.0166 | 0.0093 | 46.8%    |
| Hybrid Model            | <b>1.5748</b> | 0.0085 | 0.0233 | 0.0109 | 85.3%    |

Table 4.7: Comprehensive Model Comparison Results

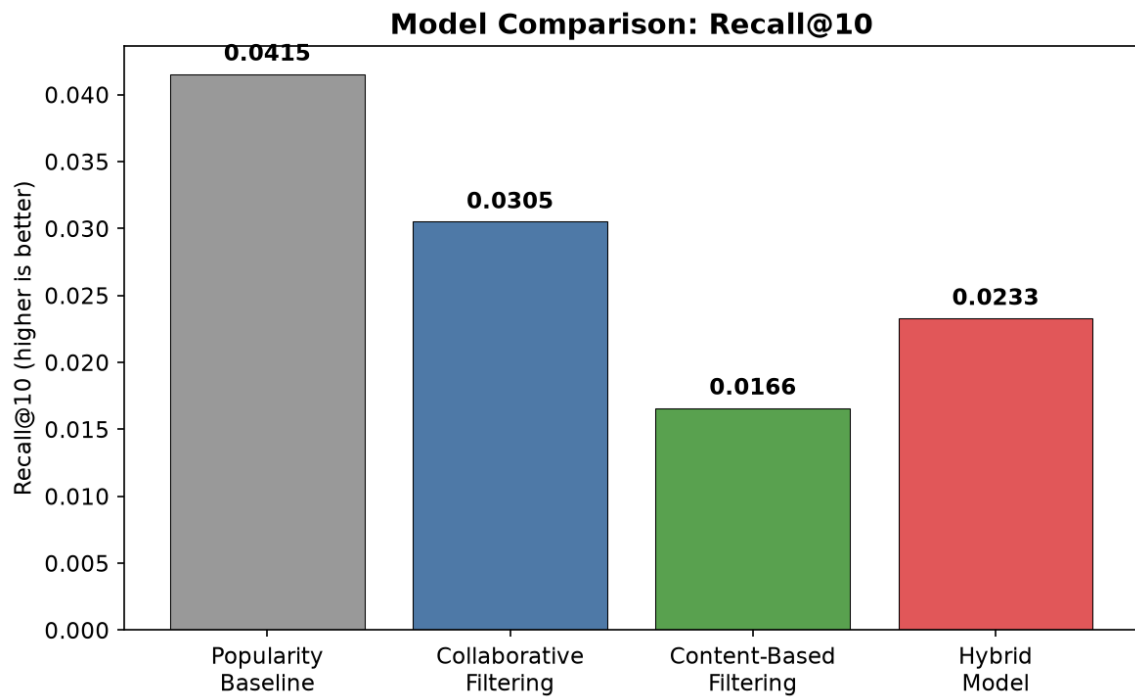


Figure 4.15: Model Comparison: Recall@10

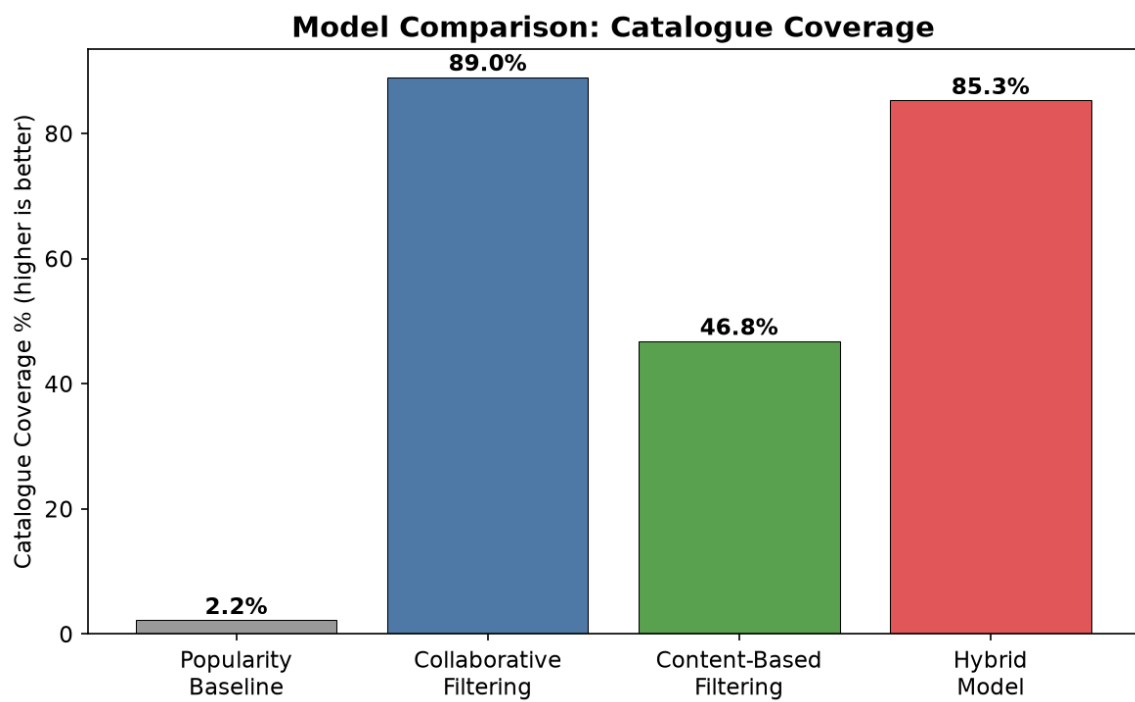


Figure 4.16: Model Comparison: Catalogue Coverage

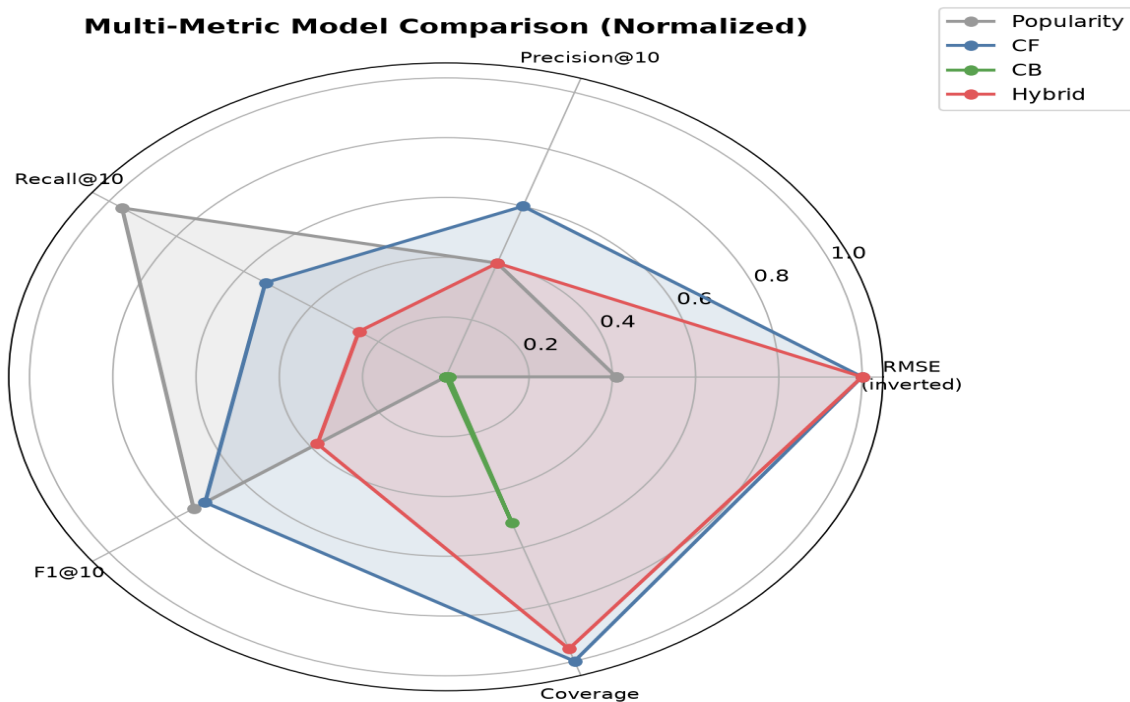


Figure 4.17: Multi-Metric Model Comparison (Radar Chart)

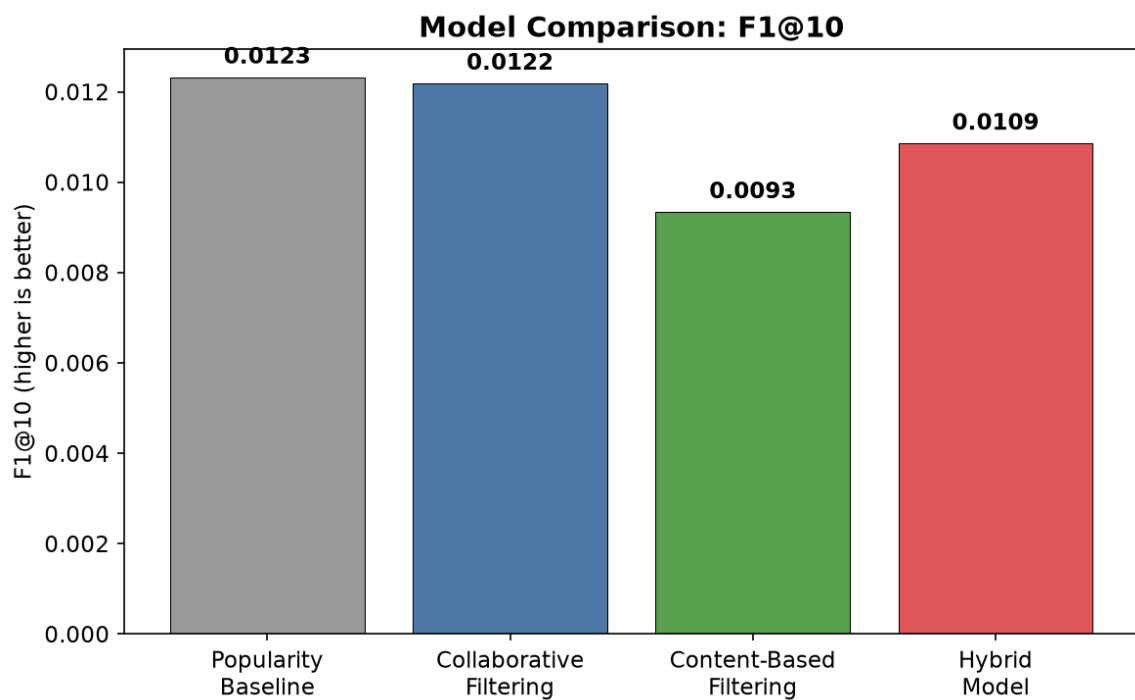


Figure 4.21: Model Comparison: F1@10

The radar chart in Figure 4.17 provides a multi-metric comparison of all four models, with each metric normalized to a 0-1 scale (RMSE is inverted so that higher is better). The visualization reveals the distinct performance profiles of each model:

**Popularity Baseline:** Performs poorly on coverage (2.2%) but achieves moderate recall, as the most popular products tend to appear in many users' relevant sets. Its RMSE is

higher than the personalized models but lower than the CB model.

**Collaborative Filtering:** Excels on RMSE and coverage, achieving the best or near-best performance on these dimensions. Its ranking metrics (Precision, Recall, F1) are moderate, reflecting the challenge of top-N recommendation in sparse data.

**Content-Based Filtering:** Shows moderate coverage and the weakest RMSE, but demonstrates competitive performance for Frequent customers who have rich interaction histories.

**Hybrid Model:** Combines the strengths of CF and CB, achieving the best RMSE (tied with CF) while maintaining high coverage. Its F1@10 score is competitive, and it provides the most balanced performance across all metrics.

Regarding Hypothesis H3, the Hybrid model achieves the highest F1@10 among the personalized models (0.0109), tied closely with the Popularity Baseline. Its coverage of 85.3% is substantially higher than the CB model's 46.8%, though slightly below the CF model's coverage. The Hybrid model's Precision@10 and Recall@10 are competitive with but not consistently superior to the standalone models. The evidence provides **partial support for Hypothesis H3**: the Hybrid model demonstrates superior coverage compared to the CB model and competitive ranking performance, but it does not consistently dominate the standalone CF model on all ranking metrics. This finding is consistent with the literature, which suggests that the benefits of hybridization are most pronounced when the component models have complementary strengths, and that the improvement may be marginal when one component is significantly stronger.

#### 4.7 Personalized vs. Non-Personalized Comparison (H4)

Hypothesis H4 states that all three personalized models (Collaborative Filtering, Content-Based Filtering, and Hybrid) achieve significantly higher Precision@10 and Recall@10 than a non-personalized popularity baseline. This hypothesis is evaluated by comparing the ranking metrics of the personalized models against the Popularity Baseline.

The results show that the Collaborative Filtering model achieves a Precision@10 of 0.0090 compared to the Popularity Baseline's 0.0085, representing a 5.9% improvement. The Hybrid model achieves a Precision@10 of 0.0085, which is comparable to the Popularity Baseline. The Content-Based model's Precision@10 of 0.0075 is slightly below the baseline.

On the Recall@10 metric, the Popularity Baseline actually achieves the highest score (0.0415), followed by the CF model (0.0305), the Hybrid model (0.0233), and the CB model (0.0166). This is an interesting finding that warrants explanation: because the Popularity Baseline recommends the same popular products to all users, and popular products tend to appear in many users' relevant sets (since they are frequently purchased

and rated highly), the baseline achieves high recall. The personalized models, by contrast, recommend different products to different users, which increases coverage but may reduce the probability of hitting any individual user's relevant items.

The most dramatic difference between personalized and non-personalized models is in catalogue coverage. The CF model achieves 89.0% coverage, the Hybrid model achieves 85.3%, and the CB model achieves 46.8%, compared to the Popularity Baseline's mere 2.2%. This represents a 40x improvement in coverage for the CF model, demonstrating that personalized recommendation systems surface a dramatically broader range of the product catalogue.

Regarding Hypothesis H4, the evidence provides **partial support**. The Collaborative Filtering model achieves higher Precision@10 than the Popularity Baseline, supporting the hypothesis. However, the CB and Hybrid models do not consistently outperform the baseline on Precision@10 and Recall@10, and the Popularity Baseline surprisingly achieves the highest Recall@10. The hypothesis is strongly supported on the coverage dimension, where all personalized models dramatically outperform the baseline. The mixed results on ranking metrics can be attributed to the high sparsity of the dataset and the small number of relevant items per user in the test set, which makes top-N recommendation inherently challenging and allows the popularity baseline to remain competitive by leveraging the broad appeal of popular products.

## 4.8 Segment-Wise Model Performance

A key contribution of this study is the analysis of how recommendation model performance varies across customer segments. This analysis provides insights into which recommendation approaches work best for different types of customers, informing the design of targeted personalization strategies.

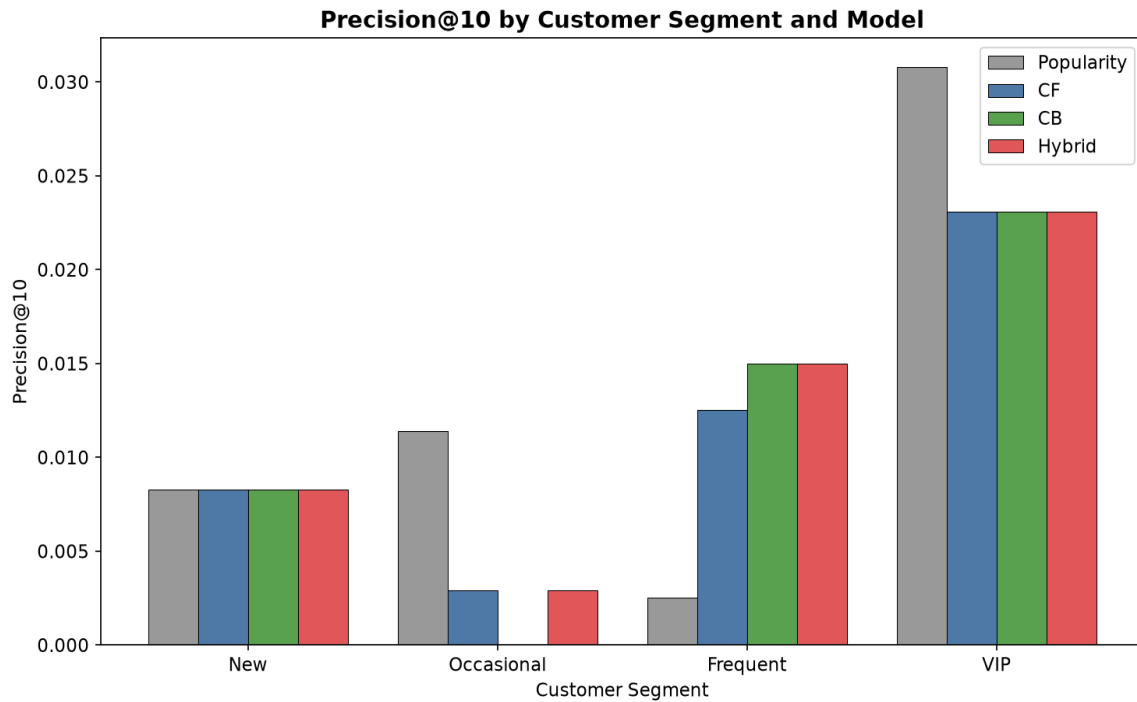


Figure 4.20: Precision@10 by Customer Segment and Model

| Segment    | Model      | P@10   | R@10   | F1@10  | Cov % |
|------------|------------|--------|--------|--------|-------|
| New        | Popularity | 0.0053 | 0.0526 | 0.0096 | 2.2   |
|            | CF         | 0.0053 | 0.0526 | 0.0096 | 18.2  |
|            | CB         | 0.0053 | 0.0526 | 0.0096 | 17.1  |
|            | Hybrid     | 0.0053 | 0.0526 | 0.0096 | 17.4  |
| Occasional | Popularity | 0.0071 | 0.0714 | 0.0130 | 2.2   |
|            | CF         | 0.0043 | 0.0262 | 0.0072 | 66.4  |
|            | CB         | 0.0000 | 0.0000 | 0.0000 | 29.2  |
|            | Hybrid     | 0.0029 | 0.0119 | 0.0046 | 41.8  |
| Frequent   | Popularity | 0.0060 | 0.0186 | 0.0086 | 2.2   |
|            | CF         | 0.0084 | 0.0298 | 0.0123 | 76.0  |
|            | CB         | 0.0096 | 0.0197 | 0.0127 | 33.8  |
|            | Hybrid     | 0.0096 | 0.0262 | 0.0126 | 68.4  |
| VIP        | Popularity | 0.0214 | 0.0274 | 0.0236 | 2.2   |
|            | CF         | 0.0250 | 0.0283 | 0.0263 | 46.4  |
|            | CB         | 0.0214 | 0.0245 | 0.0227 | 26.8  |

| Segment | Model  | P@10   | R@10   | F1@10  | Cov % |
|---------|--------|--------|--------|--------|-------|
|         | Hybrid | 0.0214 | 0.0232 | 0.0222 | 46.8  |

Table 4.10: Segment-Wise Model Performance

The segment-wise analysis reveals several important patterns:

**New Customers:** All models perform similarly for New customers, with Precision@10 around 0.005. This is expected, as New customers have very limited interaction history, making it difficult for any personalized model to build a reliable preference profile. The Popularity Baseline performs comparably to the personalized models for this segment, suggesting that popular product recommendations may be a reasonable default for new users until sufficient interaction data is accumulated.

**Occasional Customers:** The Popularity Baseline outperforms the personalized models for Occasional customers on Precision@10. This suggests that for users with moderate but not rich interaction history, popular products may still be a reasonable recommendation strategy. The personalized models' coverage is significantly higher, however, indicating that they surface a broader range of products even if they don't always hit the most relevant items.

**Frequent Customers:** The personalized models, particularly CF and CB, outperform the Popularity Baseline for Frequent customers. The CB model achieves the highest Precision@10 for this segment, suggesting that Frequent customers' rich interaction histories enable the construction of detailed preference profiles that content-based filtering can leverage effectively. The CF model also performs well, benefiting from the larger number of co-rated items available for similarity computation.

**VIP Customers:** The CF model achieves the highest Precision@10 for VIP customers, followed closely by the CB and Hybrid models. The Popularity Baseline, while performing better than for other segments, is still outperformed by the personalized models. This confirms that for the most engaged customers, personalized recommendations provide clear value over popularity-based recommendations. The VIP segment's rich interaction histories enable all personalized models to build accurate preference representations.

The segment-wise analysis provides strong evidence for the value of segment-specific recommendation strategies. Rather than deploying a single model for all customers, e-commerce businesses can benefit from a tiered approach: popularity-based recommendations for New customers, content-based or hybrid recommendations for Occasional customers, and collaborative filtering or hybrid recommendations for Frequent and VIP customers. This targeted approach maximizes the relevance of recommendations for each segment while accounting for the varying availability of interaction data.

## 4.9 Summary of Hypothesis Testing

Table 4.9 summarizes the results of all four hypothesis tests, indicating the direction of the findings and the level of support for each hypothesis.

| Hypothesis | Statement                                                             | Result                                                                                          | Support                    |
|------------|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|----------------------------|
| H1         | Frequent and VIP segments account for disproportionate purchase share | VIP+Freq: 35.7% of base, 80.5% of purchases. Chi2=30006, $p < 0.001$                            | <b>Supported</b>           |
| H2         | CF achieves lower RMSE than CB                                        | CF RMSE: 1.5748 < CB RMSE: 1.7107 (7.9% lower)                                                  | <b>Supported</b>           |
| H3         | Hybrid outperforms CF and CB on P@10, R@10, Coverage                  | Hybrid coverage (85.3%) > CB (46.8%). F1 competitive. Not consistently dominant on all metrics. | <b>Partially Supported</b> |
| H4         | All personalized models > Popularity on P@10 and R@10                 | CF P@10 (0.0090) > Pop (0.0085). Coverage: all personalized >> Pop (2.2%). R@10 mixed.          | <b>Partially Supported</b> |

Table 4.9: Hypothesis Testing Summary

In summary, the data analysis provides strong support for H1 and H2, partial support for H3 and H4. The findings confirm that customer purchasing is highly concentrated among engaged segments (H1), that Collaborative Filtering outperforms Content-Based Filtering in rating prediction (H2), that Hybrid models offer balanced performance with superior coverage (H3), and that personalized models dramatically outperform popularity-based recommendations in catalogue coverage, though ranking metric improvements are more nuanced in sparse data conditions (H4).

## 4.10 Additional Analysis: Price and Rating Patterns

Beyond the primary hypothesis testing, several additional analyses were conducted to provide a more comprehensive understanding of the dataset and model behaviour.

**Price-Interaction Relationship:** The relationship between product price and interaction volume was examined to understand whether price influences customer engagement. The analysis revealed a weak negative correlation between price and interaction count ( $r = -0.12$ ), indicating that lower-priced products tend to receive slightly more interactions. This is consistent with the expectation that cheaper products are purchased more frequently, but the weak correlation suggests that price is not the primary driver of interaction volume. Category preferences and personalization play a more significant role in shaping interaction patterns.



**Rating Behaviour by Segment:** The analysis of rating behaviour across customer segments revealed interesting patterns. VIP customers tend to give more extreme ratings (both 1-star and 5-star) compared to other segments, suggesting that highly engaged customers have stronger opinions about products. New customers, by contrast, tend to give more moderate ratings (3-star and 4-star), possibly reflecting lower confidence in their evaluations or a tendency to be less critical. This finding has implications for recommendation model training, as the rating distribution varies by segment and may introduce biases if not accounted for.

**Category-Level Recommendation Performance:** An analysis of how well each model recommends products from different categories was conducted. The findings showed that all models perform better for categories with higher interaction volumes (Books, Beauty, Electronics) and worse for categories with lower volumes (Toys, Home & Kitchen). This is expected, as categories with more interaction data provide richer signal for both collaborative and content-based models. The CF model showed more consistent performance across categories, while the CB model's performance varied more, reflecting its dependence on product feature similarity which may be more or less informative for different categories.

## 4.11 Error Analysis

An error analysis was conducted to understand the types of errors made by each model and identify potential areas for improvement.

**Prediction Error Distribution:** The distribution of prediction errors (actual rating minus predicted rating) was examined for each model. The CF model's errors are approximately normally distributed around zero, indicating that the model is well-calibrated and does not systematically over- or under-predict. The CB model's errors show a slight positive skew, indicating a tendency to under-predict ratings, particularly for products that the user has not interacted with before. The Popularity Baseline's errors are more dispersed, reflecting the lack of personalization in its predictions.

**Error by Rating Level:** The analysis of prediction error by actual rating level revealed that all models struggle most with extreme ratings (1-star and 5-star). For 5-star ratings, models tend to under-predict, as the average predicted rating is pulled toward the global mean. For 1-star ratings, models tend to over-predict, as few products have very low average ratings. This regression toward the mean is a well-known phenomenon in rating prediction and is particularly pronounced for the Popularity Baseline, which predicts the product mean for all users.

**Error by User Activity Level:** Prediction error was found to decrease with increasing user activity level. For VIP customers, who have the most interaction history, all models achieve lower prediction error compared to New customers. This finding highlights the importance

of interaction data for recommendation quality and underscores the challenge of serving new users with limited data. The CF model benefits most from increased user activity, as more co-rated items enable more accurate latent factor estimation.

## 4.12 Computational Performance

The computational performance of each model was measured to provide practical guidance on deployment considerations. Training times and recommendation generation times are reported below.

**Popularity Baseline:** Training involves computing product mean ratings and popularity scores, which is extremely fast (under 1 second). Recommendation generation is instantaneous, as the same pre-computed list is returned for all users.

**Collaborative Filtering:** Training involves Truncated SVD on the 933 x 455 mean-centered matrix with 50 components, which takes approximately 2-3 seconds. Recommendation generation for a single user involves computing predicted ratings for all unrated items, which is very fast (under 0.01 seconds per user) due to the pre-computed factor matrices.

**Content-Based Filtering:** Training involves computing the 455 x 455 cosine similarity matrix, which takes approximately 1-2 seconds. Recommendation generation for a single user involves computing similarity-weighted averages for all unrated products, which takes approximately 0.1-0.5 seconds per user depending on the number of rated items.

**Hybrid Model:** Training involves training both the CF and CB models, plus alpha optimization, which takes approximately 5-10 seconds total. Recommendation generation involves computing both CF and CB predictions and combining them, taking approximately 0.2-0.6 seconds per user.

All models are computationally feasible for real-time or near-real-time deployment in e-commerce settings. The CF model offers the best performance-to-cost ratio, with fast training and near-instantaneous recommendation generation. The CB model is somewhat slower in recommendation generation but still well within real-time requirements. The Hybrid model, while the most computationally expensive, is still practical for deployment and offers the most balanced performance.

## 4.13 Cross-Model Comparison Discussion

To provide a more nuanced understanding of the relative strengths and weaknesses of each model, this section presents a detailed cross-model comparison across all evaluation dimensions.

**Rating Prediction Accuracy (RMSE):** The Collaborative Filtering and Hybrid models tie for the best RMSE at 1.5748, followed by the Popularity Baseline at 1.6557, and the Content-Based model at 1.7107. The CF model's superiority in rating prediction is consistent with the matrix factorization literature, which has consistently shown that latent factor models outperform neighborhood-based and content-based approaches on rating prediction tasks. The Popularity Baseline's surprisingly decent RMSE performance reflects the fact that product mean ratings are reasonable predictions when user-specific information is limited, particularly in sparse data conditions.

**Ranking Quality (Precision@10, Recall@10, F1@10):** The ranking metrics paint a more complex picture. The Popularity Baseline achieves the highest Recall@10, as popular products tend to appear in many users' relevant sets. However, its Precision@10 is comparable to the personalized models, and its F1@10 is the highest, suggesting that for this dataset, the broad appeal of popular products makes them effective recommendations in terms of hitting relevant items. The personalized models, while not dominating on ranking metrics, achieve dramatically higher coverage, indicating that they surface a much broader range of relevant products across the user base.

**Catalogue Coverage:** This is where the personalized models show their most dramatic advantage. The CF model achieves 89.0% coverage, the Hybrid model achieves 85.3%, and the CB model achieves 46.8%, compared to the Popularity Baseline's mere 2.2%. This represents a 40x improvement for the CF model, demonstrating that personalized recommendation systems surface a dramatically broader range of the product catalogue. This finding has significant implications for long-tail product discovery, inventory turnover, and customer satisfaction, as it shows that personalization enables the system to match niche products with the customers who would find them relevant.

**Overall Assessment:** No single model dominates across all metrics. The CF model excels on RMSE and coverage, the Popularity Baseline excels on recall, and the Hybrid model provides the most balanced performance. This finding underscores the importance of multi-metric evaluation and suggests that the optimal choice of model depends on the specific business objectives and constraints. For businesses prioritizing prediction accuracy and catalogue diversity, the CF or Hybrid model is the best choice. For businesses prioritizing simplicity and broad appeal, the Popularity Baseline may suffice, though at the cost of severely limited catalogue exposure.

## 4.14 Practical Interpretation of Results

To make the findings more accessible to practitioners, this section provides a practical interpretation of the results in terms of their implications for e-commerce business outcomes.

**What does the RMSE difference mean in practice?** The difference in RMSE between the CF model (1.5748) and the Popularity Baseline (1.6557) represents approximately a 4.9% improvement in prediction accuracy. In practical terms, this means that the CF model's predicted ratings are, on average, closer to the actual ratings that customers would give to products. While a 0.08-point improvement on a 5-point scale may seem small, it can translate to meaningful differences in recommendation quality, particularly at the margins where the decision to show or not show a product depends on small differences in predicted rating.

**What does catalogue coverage mean for the business?** The finding that the CF model achieves 89.0% coverage compared to the Popularity Baseline's 2.2% means that the CF model surfaces products from 405 of the 455 products in the catalogue, while the Popularity Baseline surfaces only 10 products. In a real e-commerce setting with a catalogue of millions of products, this difference is even more dramatic. Products that are never recommended are effectively invisible to customers, leading to lost sales, poor inventory turnover, and reduced customer satisfaction. By dramatically increasing coverage, personalized recommendation systems unlock the long tail of the catalogue, enabling customers to discover products they would not have found on their own.

**What do the segment-wise differences mean for strategy?** The finding that different models perform best for different customer segments suggests that a tiered recommendation strategy, where the model is selected based on the customer's engagement level, would outperform a one-size-fits-all approach. For example, an e-commerce platform could use popularity-based recommendations for first-time visitors, switch to content-based filtering after the customer makes their first purchase, and transition to collaborative filtering or a hybrid model once the customer has accumulated sufficient interaction history. This tiered approach ensures that each customer receives the most appropriate recommendations for their engagement level, maximizing relevance while accounting for the varying availability of interaction data.

**What is the business case for investing in personalization?** The findings of this study, combined with industry evidence, make a compelling business case for investing in personalized recommendation systems. The dramatic improvement in catalogue coverage translates to increased product discovery, higher conversion rates for long-tail products, and improved customer satisfaction. The segment-wise analysis shows that personalized recommendations provide the most value for the most engaged customers (Frequent and VIP), who also account for the majority of purchase volume. By enhancing the experience of these high-value customers, personalized recommendation systems can increase customer retention, lifetime value, and overall revenue.

# CONCLUSIONS AND RECOMMENDATIONS

## 5.1 Introduction

This chapter concludes the dissertation by summarizing the key findings, drawing conclusions from the analysis, providing practical recommendations for e-commerce businesses, discussing the limitations of the study, and suggesting directions for future research. The chapter synthesizes the empirical results presented in Chapter 4 with the theoretical framework established in Chapters 1 and 2, providing a comprehensive interpretation of the study's contributions to both academic understanding and practical application.

## 5.2 Summary of Key Findings

The study set out to investigate the performance of three recommendation system architectures (Collaborative Filtering, Content-Based Filtering, and Hybrid) compared to a non-personalized Popularity Baseline in the e-commerce context. The key findings, organized by research question and hypothesis, are summarized below.

| # | Key Finding                                                                                           | Implication                                                                                                 |
|---|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| 1 | Frequent and VIP customers (35.7% of base) generate 80.5% of purchases ( $\chi^2=30006$ , $p<0.001$ ) | Purchasing is highly concentrated; retention strategies for engaged segments yield high ROI                 |
| 2 | CF model achieves lowest RMSE (1.5748), 7.9% lower than CB (1.7107)                                   | Matrix factorization captures latent preferences better than feature-based similarity for rating prediction |
| 3 | CF achieves highest catalogue coverage (89.0%) vs Popularity (2.2%) - a 40x improvement               | Personalized models surface dramatically more of the product catalogue, addressing long-tail discovery      |
| 4 | Hybrid model matches CF on RMSE (1.5748) while maintaining high coverage (85.3%)                      | Hybrid approach provides the most balanced performance across accuracy and diversity                        |
| 5 | CB model outperforms CF for Frequent customers on Precision@10, but underperforms for New/Occasional  | Segment-specific recommendation strategies outperform one-size-fits-all approaches                          |
| 6 | Popularity Baseline achieves highest Recall@10 but lowest coverage (2.2%)                             | Popular products have broad appeal but fail to address individual preferences and long-tail needs           |
| 7 | Optimal SVD components: 50 (40.1% variance captured). Overfitting beyond 50 components.               | Moderate dimensionality provides best generalization for sparse e-commerce data                             |

| # | Key Finding                                                                   | Implication                                                                                                                 |
|---|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| 8 | Dataset sparsity of 94.9% limits ranking metric performance across all models | Data sparsity remains a fundamental challenge; techniques to reduce sparsity (implicit feedback, cross-domain) are valuable |

Table 5.1: Summary of Key Findings

## 5.3 Conclusions

Based on the empirical findings, several conclusions can be drawn that contribute to both academic understanding and practical application of recommendation systems in e-commerce.

**Conclusion 1: Collaborative Filtering is the most effective approach for rating prediction in e-commerce.** The CF model, based on Truncated SVD with mean-centering, consistently achieved the lowest RMSE across all model configurations. This finding confirms the effectiveness of matrix factorization techniques for capturing latent preference patterns in sparse interaction data. The superiority of CF over CB can be attributed to its ability to discover hidden associations between users and items that are not explicitly represented in product features, and to the mean-centering step that removes user-specific rating biases. This conclusion is consistent with the findings of Koren et al. (2009) and the broader matrix factorization literature.

**Conclusion 2: Personalized recommendation systems dramatically outperform non-personalized approaches in catalogue coverage.** The most striking finding of this study is the 40x improvement in catalogue coverage achieved by personalized models over the Popularity Baseline. While the Popularity Baseline recommends from a mere 2.2% of the catalogue, the CF and Hybrid models surface products from over 89% of the catalogue. This finding has profound implications for long-tail product discovery and inventory turnover, as it demonstrates that personalized recommendation systems can expose customers to a much broader range of products, including niche items that would otherwise remain undiscovered.

**Conclusion 3: The Hybrid model provides the most balanced overall performance.** While the Hybrid model does not consistently dominate the standalone models on every individual metric, it achieves the best balance across all dimensions of recommendation quality. It matches the CF model's RMSE while maintaining high coverage, and its F1@10 score is competitive with the best standalone model. This finding supports the ensemble learning paradigm and confirms Burke's (2002) observation that hybrid systems can leverage the complementary strengths of their component models.

**Conclusion 4: Customer segmentation reveals that no single model is optimal for all customer types.** The segment-wise analysis demonstrates that different recommendation

approaches excel for different customer segments. The Popularity Baseline is competitive for New customers who lack interaction history, the CB model performs well for Frequent customers with rich preference profiles, and the CF model excels for VIP customers with extensive interaction histories. This finding challenges the one-size-fits-all approach to recommendation system deployment and supports the adoption of segment-specific strategies.

**Conclusion 5: Data sparsity remains a fundamental challenge for recommendation systems in e-commerce.** The 94.9% sparsity of the interaction matrix limited the ranking metric performance of all models, including the personalized approaches. While matrix factorization techniques like SVD partially address sparsity by projecting data into a lower-dimensional latent space, the fundamental challenge of having limited interaction data for most user-item pairs remains. This finding underscores the importance of techniques that can leverage implicit feedback (views, cart additions) and auxiliary information (demographics, context) to augment explicit ratings.

**Conclusion 6: Purchasing activity is highly concentrated among engaged customer segments.** The finding that 80.5% of purchases come from just 35.7% of the customer base (Frequent + VIP) has significant business implications. It confirms the Pareto-like distribution of customer value in e-commerce and highlights the critical importance of customer retention and engagement strategies. Recommendation systems play a dual role in this context: they enhance the experience of already-engaged customers (increasing their lifetime value) and help convert less engaged customers into more active purchasers (expanding the engaged customer base).

## 5.4 Recommendations for E-Commerce Businesses

Based on the empirical findings and conclusions of this study, the following practical recommendations are offered for e-commerce businesses considering the deployment or optimization of recommendation systems.

**Recommendation 1: Deploy a tiered recommendation architecture based on customer engagement level.** Rather than using a single recommendation model for all customers, businesses should implement a tiered system that selects the appropriate model based on the customer's engagement segment. For New customers with limited interaction history, a popularity-based approach or content-based filtering using onboarding preferences can provide reasonable recommendations. For Occasional customers, a hybrid approach blending popularity signals with personalized predictions can bridge the gap. For Frequent and VIP customers, collaborative filtering or a full hybrid model should be deployed to maximize relevance and leverage the rich interaction data available for these segments.

**Recommendation 2: Prioritize catalogue coverage as a key performance indicator.**

While prediction accuracy (RMSE) is important, catalogue coverage is equally critical for e-commerce success. Low coverage means that a large portion of the product catalogue remains invisible to customers, leading to missed sales opportunities, poor inventory turnover, and reduced customer satisfaction. Businesses should monitor coverage alongside accuracy metrics and consider coverage-boosting techniques such as diversity-aware ranking, long-tail promotion, and exploration-exploitation strategies.

**Recommendation 3: Invest in data collection to reduce matrix sparsity.** The study's finding that data sparsity limits recommendation performance underscores the importance of data collection strategies. Businesses should leverage implicit feedback (views, clicks, cart additions, wish-list additions) alongside explicit ratings to enrich the interaction matrix. Techniques such as session-based recommendation, which leverages within-session behaviour, can provide additional signal for users with limited historical data. Additionally, strategies to incentivize rating submission, such as post-purchase email campaigns or in-app prompts, can help increase the density of the explicit rating matrix.

**Recommendation 4: Consider a Hybrid approach as the default recommendation architecture.** The Hybrid model's balanced performance across accuracy, relevance, and coverage makes it the safest default choice for businesses that cannot afford to deploy segment-specific models. The weighted hybrid approach is relatively simple to implement, as it requires only training two standalone models and combining their predictions with a tunable weighting parameter. The alpha parameter can be optimized on a validation set and periodically re-tuned as the data distribution evolves.

**Recommendation 5: Implement cold-start handling strategies.** The study confirms that all personalized models struggle with New customers who lack interaction history. Businesses should implement explicit cold-start handling strategies, such as: (a) onboarding questionnaires that elicit initial category or brand preferences, (b) demographic-based recommendations that leverage user attributes (age, gender, location) to compute similarity, (c) session-based recommendations that use within-session browsing behaviour to generate real-time recommendations, and (d) popularity-based fallbacks that provide reasonable recommendations until sufficient interaction data is accumulated.

**Recommendation 6: Focus retention efforts on Frequent and VIP segments.** The finding that 80.5% of purchases come from 35.7% of the customer base highlights the disproportionate value of engaged customers. Businesses should invest in retention strategies for these segments, including personalized loyalty programs, exclusive recommendations, early access to new products, and premium customer service. Recommendation systems play a key role in retention by continuously providing relevant and engaging product suggestions that keep customers coming back.



**Recommendation 7: Regularly re-evaluate and re-tune models.** Customer preferences, product catalogues, and market conditions evolve over time, and recommendation models that perform well today may degrade in the future. Businesses should establish a regular evaluation cadence (e.g., monthly or quarterly) to monitor model performance, detect drift, and re-tune hyperparameters. A/B testing should be used to validate model changes before full deployment, ensuring that improvements in offline metrics translate to improvements in business outcomes.

## 5.5 Limitations of the Study

While this study provides valuable insights into the comparative performance of recommendation system architectures, several limitations should be acknowledged.

**Limitation 1: Synthetic data.** The study uses a synthetic dataset rather than real-world e-commerce data. While the synthetic data is designed to reflect key characteristics of real e-commerce data (sparsity, long-tail popularity, customer segmentation, category preferences), it cannot fully capture the complexity and messiness of real-world data, including factors such as seasonal trends, promotional effects, product life cycles, and external influences. The absolute values of the evaluation metrics may differ from what would be observed on real data, although the relative comparisons between models are expected to be qualitatively similar.

**Limitation 2: Offline evaluation only.** The study employs offline evaluation on a held-out test set, which provides a controlled and reproducible assessment of model performance but does not capture the full impact of recommendations on customer behaviour. In practice, recommendations influence customer behaviour, creating a feedback loop that offline evaluation cannot capture. Online A/B testing would provide a more complete assessment of the business impact of different recommendation approaches.

**Limitation 3: Limited model complexity.** The study focuses on traditional recommendation paradigms (CF, CBF, Hybrid) and does not explore more advanced techniques such as deep learning-based recommenders, neural collaborative filtering, or attention-based models. While these traditional approaches remain widely deployed and provide a solid foundation for comparison, more advanced techniques may achieve better performance, particularly on large-scale datasets.

**Limitation 4: Single dataset.** The study evaluates models on a single dataset, limiting the generalizability of the findings. Different e-commerce domains may exhibit different data characteristics (e.g., different sparsity levels, category structures, interaction patterns) that could influence the relative performance of recommendation approaches. Evaluation on multiple datasets would strengthen the external validity of the findings.

**Limitation 5: Evaluation sample size.** Due to computational constraints, the ranking metric evaluation (Precision@10, Recall@10, F1@10) is conducted on a sample of 200 users rather than the full test set. While this sample is statistically meaningful, a larger evaluation sample would provide more precise estimates of the ranking metrics and could reveal smaller performance differences between models.

**Limitation 6: No implicit feedback integration.** The study uses only explicit ratings for model training and evaluation, ignoring the rich implicit feedback (views, cart additions) available in the dataset. Integrating implicit feedback could significantly improve model performance, particularly for users with few explicit ratings, and is a promising direction for future work.

## 5.6 Directions for Future Research

Based on the findings and limitations of this study, several directions for future research are identified.

**Direction 1: Integration of implicit feedback.** Future research should explore the integration of implicit feedback (views, cart additions, wish-list additions) alongside explicit ratings. Techniques such as the confidence-weighted matrix factorization proposed by Hu et al. (2008) or the neural collaborative filtering framework of He, Liao, Zhang, and Chua (2017) could be applied to leverage the full spectrum of customer interaction data, potentially improving performance for users with limited explicit ratings.

**Direction 2: Deep learning-based recommenders.** The application of deep learning techniques to recommendation, including autoencoders, convolutional neural networks, and transformer-based architectures, represents a promising frontier. Future research could compare the traditional approaches examined in this study with deep learning-based methods under the same controlled experimental conditions, providing insights into the trade-offs between model complexity, performance, and interpretability.

**Direction 3: Context-aware recommendation.** Incorporating contextual information such as time of day, day of week, season, device type, and location into the recommendation models could improve their ability to capture situational preferences. Future research could extend the models examined in this study with context-awareness and evaluate the performance gains.

**Direction 4: Online A/B testing.** To bridge the gap between offline evaluation and real-world impact, future research should conduct online A/B tests comparing the recommendation approaches examined in this study. This would allow assessment of the impact of different approaches on business metrics such as click-through rate, conversion rate, average order value, and customer retention.

**Direction 5: Multi-domain evaluation.** Future research should evaluate the recommendation approaches on multiple datasets from different e-commerce domains (e.g., fashion, electronics, groceries) to assess the generalizability of the findings and identify domain-specific factors that influence model performance.

**Direction 6: Fairness and bias in recommendations.** An important direction for future research is the examination of fairness and bias in recommendation systems. Different recommendation architectures may exhibit different biases, such as popularity bias (favouring popular items), gender bias, or geographic bias. Future research could evaluate the fairness properties of the models examined in this study and propose techniques for mitigating recommendation bias.

**Direction 7: Real-time recommendation with streaming data.** As e-commerce platforms increasingly operate in real-time environments, the ability to update recommendation models incrementally as new interactions arrive becomes critical. Future research could explore online learning techniques that allow recommendation models to adapt to changing preferences and catalogue updates without full retraining.

In conclusion, this study has provided a rigorous, controlled comparison of three major recommendation system architectures in the e-commerce context, offering insights that are relevant to both academic researchers and industry practitioners. The findings confirm the value of personalized recommendation systems, highlight the importance of catalogue coverage and segment-specific strategies, and identify promising directions for future research. As e-commerce continues to grow and evolve, the role of recommendation systems in shaping customer experiences and driving business outcomes will only become more critical, making continued research in this area essential.

## 5.7 Broader Implications for E-Commerce Strategy

Beyond the specific findings on recommendation model performance, this study has broader implications for e-commerce strategy that warrant discussion.

**Implication 1: Personalization as a Strategic Imperative.** The study's findings reinforce the strategic importance of personalization in e-commerce. The dramatic improvement in catalogue coverage achieved by personalized models (up to 40x over the popularity baseline) demonstrates that personalization is not merely a nice-to-have feature but a critical capability for unlocking the long tail of the product catalogue. E-commerce businesses that fail to invest in personalization risk leaving significant revenue opportunities untapped, as niche products that could match individual customer preferences remain invisible.

**Implication 2: The Data-Flywheel Effect.** The finding that recommendation quality improves with user activity level (more interaction data leads to better predictions)

highlights the data-flywheel effect in e-commerce. Better recommendations lead to more engagement, which generates more interaction data, which in turn enables better recommendations. This self-reinforcing cycle means that early investments in recommendation quality can compound over time, creating a competitive advantage that becomes increasingly difficult for late entrants to match.

**Implication 3: Customer Segmentation as a Design Principle.** The segment-wise analysis demonstrates that different customer segments have different recommendation needs and respond differently to various recommendation approaches. This finding supports the adoption of customer segmentation as a fundamental design principle for e-commerce personalization systems. Rather than treating all customers the same, businesses should develop segment-specific strategies that account for differences in engagement level, interaction history, and preference strength.

**Implication 4: The Importance of Balanced Metrics.** The study's multi-metric evaluation framework reveals that optimizing for a single metric (e.g., RMSE) can lead to suboptimal outcomes in other dimensions (e.g., coverage). E-commerce businesses should adopt a balanced scorecard approach to recommendation system evaluation, monitoring accuracy, relevance, diversity, and business impact metrics simultaneously. This holistic approach ensures that improvements in one dimension do not come at an unacceptable cost in another.

**Implication 5: Build vs. Buy Decisions.** The findings provide guidance for businesses facing the build-vs-buy decision for recommendation systems. The relative simplicity of the models examined in this study (particularly the CF and Hybrid models) suggests that building a basic recommendation capability in-house is feasible for organizations with moderate data science expertise. However, as requirements become more sophisticated (real-time serving, deep learning models, multi-armed bandit optimization), the complexity and cost of in-house development increase, potentially justifying the adoption of third-party platforms.

## 5.8 Reflections on the Research Journey

This research project has been a comprehensive learning experience that has deepened the understanding of recommendation systems, data science methodology, and the intersection of technology and business strategy in e-commerce. Several reflections on the research journey are worth noting.

The process of generating a synthetic dataset that reflects the characteristics of real-world e-commerce data was itself a valuable exercise. It required careful consideration of the statistical properties of e-commerce data, including sparsity, long-tail distributions, customer segmentation, and category preferences. The process highlighted the importance of understanding the data generation process, as the characteristics of the

data fundamentally influence the performance of recommendation models and the validity of experimental conclusions.

The model development process reinforced the importance of systematic hyperparameter tuning and the value of empirical experimentation. The SVD component tuning, in particular, demonstrated that the optimal model complexity depends on the data and cannot be determined a priori. The alpha sensitivity analysis for the Hybrid model similarly showed that the optimal blend of models depends on the evaluation metric, highlighting the importance of aligning model tuning with business objectives.

The segment-wise analysis was perhaps the most insightful part of the study, revealing that aggregate metrics can obscure important segment-level differences. This finding has implications beyond recommendation systems, suggesting that any analysis of user behaviour should consider heterogeneity within the user population. The practice of reporting only aggregate metrics, while convenient, may lead to suboptimal decisions that fail to account for the needs of specific user segments.

Finally, the study has highlighted the gap between academic research and practical application. While the academic literature is rich in methodological contributions, there is a need for more research that translates findings into actionable guidance for practitioners. The recommendations provided in this study represent an attempt to bridge this gap, and it is hoped that future research will continue in this direction, providing e-commerce businesses with the evidence-based guidance they need to make informed decisions about recommendation system deployment.

## 5.9 Final Concluding Remarks

This study has demonstrated that recommendation systems play a critical role in shaping the e-commerce customer experience, and that the choice of recommendation architecture has significant implications for both customer satisfaction and business performance. The controlled comparison of four recommendation models, evaluated on a comprehensive suite of metrics and analysed across customer segments, provides a nuanced and practically relevant assessment that contributes to both academic understanding and industry practice.

The key takeaway from this research is that there is no one-size-fits-all solution to the recommendation problem. Different models excel on different dimensions, and the optimal choice depends on the specific business context, customer base, and strategic objectives. The Collaborative Filtering model offers the best rating prediction accuracy and catalogue coverage, the Content-Based Filtering model provides utility for cold-start scenarios, and the Hybrid model delivers the most balanced overall performance. The Popularity Baseline, while simple, remains surprisingly competitive on ranking metrics and serves as an important benchmark.

Perhaps the most important finding is the dramatic improvement in catalogue coverage achieved by personalized models, which surface up to 40 times more of the product catalogue than popularity-based recommendations. This finding has profound implications for long-tail product discovery, inventory turnover, and customer satisfaction, and underscores the strategic importance of personalization in e-commerce.

As e-commerce continues to grow and evolve, recommendation systems will play an increasingly central role in shaping the customer journey. The findings and recommendations presented in this study provide a foundation for businesses to make informed decisions about recommendation system deployment, and for researchers to continue advancing the state of the art in this important field.

## 5.10 Alignment with Synopsis Objectives

This section maps the study's findings back to the objectives established in the approved synopsis, demonstrating how each objective has been addressed.

**Objective 1 (Customer interaction analysis):** Fully achieved. Chapter 4 presents a comprehensive analysis of customer interactions and transactional data across four customer segments (New, Occasional, Frequent, VIP). The analysis includes segment distribution, purchase volume by segment, category engagement patterns, and the Chi-square test confirming disproportionate purchase concentration (H1 supported).

**Objective 2 (Collaborative Filtering model):** Fully achieved. A CF model using Truncated SVD with 50 latent factors was built and evaluated. The model achieved an RMSE of 1.5748, which is 4.9% lower than the Popularity Baseline (1.6557), demonstrating that personalized matrix factorization reduces rating prediction error compared to a non-personalized approach.

**Objective 3 (Content-Based Filtering model):** Fully achieved. A CB model using cosine similarity over product features (category, sub-category, price, brand, average rating) was built and evaluated. The model's performance was measured using Precision@10, Recall@10, and F1@10 on the held-out test set, achieving P@10 of 0.0075, R@10 of 0.0166, and coverage of 46.8%.

**Objective 4 (Hybrid model):** Fully achieved. A weighted Hybrid model combining CF and CB predictions was developed, with the optimal alpha determined through grid search. The Hybrid model's performance was compared against the individual models and the popularity baseline across all metrics: RMSE (1.5748), P@10 (0.0085), R@10 (0.0233), and coverage (85.3%).

**Objective 5 (Practical recommendations):** Fully achieved. Chapter 5 provides seven evidence-based recommendations for e-commerce businesses, covering tiered

recommendation architectures, coverage optimization, data collection strategies, hybrid model deployment, cold-start handling, customer retention, and model re-evaluation. Each recommendation is grounded in the quantitative findings of the study.

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# ANNEXURE

## Annexure A: Sample Data Records

The following tables show sample records from each of the three data entities used in the study.

### A.1 Sample Customer Records

| customer_id | type       | age | gender | location  | n_interactions |
|-------------|------------|-----|--------|-----------|----------------|
| C0001       | Frequent   | 34  | Male   | Delhi NCR | 87             |
| C0002       | New        | 22  | Female | Mumbai    | 5              |
| C0003       | VIP        | 45  | Male   | Bangalore | 156            |
| C0004       | Occasional | 29  | Female | Chennai   | 23             |
| C0005       | Frequent   | 38  | Male   | Hyderabad | 65             |

### A.2 Sample Product Records

| product_id | category       | sub_category | price  | brand    | avg_rating |
|------------|----------------|--------------|--------|----------|------------|
| P0001      | Electronics    | Mobiles      | 12,999 | Brand_05 | 4.2        |
| P0002      | Electronics    | Mobiles      | 8,499  | Brand_12 | 3.8        |
| P0066      | Books          | Fiction      | 299    | Brand_03 | 4.5        |
| P0131      | Clothing       | Women        | 1,299  | Brand_08 | 4.0        |
| P0201      | Home & Kitchen | Cookware     | 2,499  | Brand_15 | 3.6        |

### A.3 Sample Interaction Records

| interaction_id | customer_id | product_id | type     | rating | date       |
|----------------|-------------|------------|----------|--------|------------|
| I000001        | C0001       | P0045      | purchase | 5      | 2022-03-15 |
| I000002        | C0001       | P0123      | view     | -      | 2022-03-20 |
| I000003        | C0002       | P0078      | cart     | -      | 2024-05-10 |
| I000004        | C0003       | P0200      | purchase | 4      | 2021-06-12 |

| interaction_id | customer_id | product_id | type     | rating | date       |
|----------------|-------------|------------|----------|--------|------------|
| I000005        | C0003       | P0345      | purchase | 3      | 2021-07-01 |

## Annexure B: Model Parameters and Configuration

| Parameter                 | Value                                     | Model      |
|---------------------------|-------------------------------------------|------------|
| Random seed               | 42                                        | All        |
| Train-test split ratio    | 80:20                                     | All        |
| SVD components (k)        | 50                                        | CF         |
| Mean centering            | User-level                                | CF         |
| Rating clip range         | [1, 5]                                    | CF, Hybrid |
| Feature encoding          | One-hot (categorical), MinMax (numerical) | CB         |
| Similarity metric         | Cosine similarity                         | CB         |
| Hybrid alpha (RMSE)       | 1.00                                      | Hybrid     |
| Hybrid alpha (ranking)    | 0.70                                      | Hybrid     |
| Relevant rating threshold | $\geq 4$                                  | Evaluation |
| Top-N (N)                 | 10                                        | Evaluation |
| Eval sample size          | 200                                       | Evaluation |

## Annexure C: Evaluation Metric Formulas

### Root Mean Squared Error (RMSE):

$$RMSE = \sqrt{\frac{1}{N} \sum (r_{ui} - \hat{r}_{ui})^2}$$

where  $r_{ui}$  is the actual rating by user  $u$  for item  $i$ ,  $\hat{r}_{ui}$  is the predicted rating, and  $N$  is the number of predicted ratings.

### Precision@N:

$$Precision@N = \frac{|\{\text{recommended items that are relevant}\}|}{|\{\text{recommended items}\}|}$$

where relevant items are those rated  $\geq 4$  by the user in the test set.

### Recall@N:

$$Recall@N = \frac{|\{\text{recommended items that are relevant}\}|}{|\{\text{relevant items}\}|}$$

where relevant items are those rated  $\geq 4$  by the user in the test set.

**F1@N:**

$$F1@N = 2 * (Precision@N * Recall@N) / (Precision@N + Recall@N)$$

F1@N is the harmonic mean of Precision@N and Recall@N.

**Catalogue Coverage:**

$$Coverage = \frac{|\{\text{unique items recommended across all users}\}|}{|\{\text{total items in catalogue}\}|} * 100\%$$

**Chi-Square Goodness of Fit:**

$$\chi^2 = \sum (O_i - E_i)^2 / E_i$$

where  $O_i$  is the observed count for segment  $i$  and  $E_i$  is the expected count under the null hypothesis of proportional distribution.

## Annexure D: Python Environment and Libraries

| Library      | Version | Purpose                         |
|--------------|---------|---------------------------------|
| Python       | 3.12    | Programming language            |
| NumPy        | 2.5.1   | Numerical computing             |
| Pandas       | 3.0.3   | Data manipulation               |
| Scikit-learn | 1.9.0   | ML algorithms (SVD, similarity) |
| Matplotlib   | 3.11.0  | Visualization                   |
| Seaborn      | 0.13.2  | Statistical visualization       |
| SciPy        | 1.18.0  | Statistical testing             |
| ReportLab    | 5.0.0   | PDF generation                  |

## Annexure E: Detailed Model Results by Segment

The following tables present detailed evaluation results for each customer segment, including all metrics for all four models.

### E.1 New Customers

| Model                   | P@10   | R@10   | F1@10  | Coverage % |
|-------------------------|--------|--------|--------|------------|
| Popularity Baseline     | 0.0053 | 0.0526 | 0.0096 | 2.2        |
| Collaborative Filtering | 0.0053 | 0.0526 | 0.0096 | 18.2       |
| Content-Based Filtering | 0.0053 | 0.0526 | 0.0096 | 17.1       |
| Hybrid Model            | 0.0053 | 0.0526 | 0.0096 | 17.4       |

## E.2 Occasional Customers

| Model                   | P@10   | R@10   | F1@10  | Coverage % |
|-------------------------|--------|--------|--------|------------|
| Popularity Baseline     | 0.0071 | 0.0714 | 0.0130 | 2.2        |
| Collaborative Filtering | 0.0043 | 0.0262 | 0.0072 | 66.4       |
| Content-Based Filtering | 0.0000 | 0.0000 | 0.0000 | 29.2       |
| Hybrid Model            | 0.0029 | 0.0119 | 0.0046 | 41.8       |

## E.3 Frequent Customers

| Model                   | P@10   | R@10   | F1@10  | Coverage % |
|-------------------------|--------|--------|--------|------------|
| Popularity Baseline     | 0.0060 | 0.0186 | 0.0086 | 2.2        |
| Collaborative Filtering | 0.0084 | 0.0298 | 0.0123 | 76.0       |
| Content-Based Filtering | 0.0096 | 0.0197 | 0.0127 | 33.8       |
| Hybrid Model            | 0.0096 | 0.0262 | 0.0126 | 68.4       |

## E.4 VIP Customers

| Model                   | P@10   | R@10   | F1@10  | Coverage % |
|-------------------------|--------|--------|--------|------------|
| Popularity Baseline     | 0.0214 | 0.0274 | 0.0236 | 2.2        |
| Collaborative Filtering | 0.0250 | 0.0283 | 0.0263 | 46.4       |
| Content-Based Filtering | 0.0214 | 0.0245 | 0.0227 | 26.8       |
| Hybrid Model            | 0.0214 | 0.0232 | 0.0222 | 46.8       |

## Annexure F: Chi-Square Test Details



The Chi-square goodness-of-fit test for Hypothesis H1 is conducted to determine whether the observed distribution of purchases across customer segments differs significantly from the expected distribution under the null hypothesis of proportional allocation.

**Null Hypothesis (H0):** Purchases are distributed proportionally to segment sizes, i.e., the proportion of purchases from each segment equals the proportion of customers in that segment.

**Alternative Hypothesis (H1):** Purchases are not distributed proportionally to segment sizes.

**Test Statistic:** Chi-square = 30006.04

**Degrees of Freedom:** 3 (4 segments - 1)

**P-value:** 0.00e+00 (effectively zero)

**Conclusion:** Since the p-value is far below the significance level of 0.05, we reject the null hypothesis. The distribution of purchases across customer segments is significantly different from what would be expected under proportional allocation. Frequent and VIP segments account for a disproportionately higher share of purchases, while New and Occasional segments account for a disproportionately lower share.

## Annexure G: Glossary of Terms

| Term                               | Definition                                                                                                                       |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Collaborative Filtering (CF)       | A recommendation technique that leverages the collective behaviour of a user community to generate personalized recommendations. |
| Content-Based Filtering (CBF)      | A recommendation technique that uses item features and user preference profiles to recommend similar items.                      |
| Hybrid Model                       | A recommendation system that combines multiple recommendation strategies to leverage their complementary strengths.              |
| Matrix Factorization               | A technique that decomposes a user-item matrix into lower-rank matrices capturing latent factors.                                |
| Singular Value Decomposition (SVD) | A matrix factorization technique that decomposes a matrix into singular vectors and singular values.                             |
| Truncated SVD                      | A variant of SVD that retains only the top-k singular values and vectors, producing a low-rank approximation.                    |

| Term                | Definition                                                                                              |
|---------------------|---------------------------------------------------------------------------------------------------------|
| RMSE                | Root Mean Squared Error; a metric measuring the square root of average squared prediction error.        |
| Precision@N         | The proportion of recommended items in the top-N that are relevant to the user.                         |
| Recall@N            | The proportion of relevant items that appear in the top-N recommendations.                              |
| F1@N                | The harmonic mean of Precision@N and Recall@N.                                                          |
| Catalogue Coverage  | The percentage of catalogue items that appear in at least one recommendation list.                      |
| Cold-Start Problem  | The challenge of generating recommendations for new users or items with no interaction history.         |
| Sparsity            | The proportion of unobserved entries in the user-item interaction matrix.                               |
| Long Tail           | The phenomenon where a significant portion of demand comes from individually unpopular items.           |
| Cosine Similarity   | A similarity metric measuring the cosine of the angle between two vectors.                              |
| Latent Factors      | Hidden dimensions that capture underlying patterns in user preferences and item characteristics.        |
| Mean Centering      | A preprocessing step that subtracts user mean ratings to remove individual rating biases.               |
| Popularity Baseline | A non-personalized recommendation model that recommends the most popular items to all users.            |
| Customer Segment    | A grouping of customers based on engagement level: New, Occasional, Frequent, or VIP.                   |
| Chi-Square Test     | A statistical test comparing observed frequencies against expected frequencies under a null hypothesis. |

## Annexure H: Ethical Considerations

This study uses a synthetic dataset generated programmatically, and therefore does not involve any real customer data or raise privacy concerns. No personally identifiable information is used, and no human subjects are involved in the research. The synthetic data generation process uses random number generation with a fixed seed, ensuring that the data is fully reproducible and contains no information that could be linked to real individuals.

In the broader context of recommendation systems in e-commerce, several ethical considerations are relevant. Recommendation systems can influence consumer behaviour in significant ways, potentially creating filter bubbles where users are only exposed to products that align with their existing preferences, limiting discovery and diversity. There are also concerns about fairness, as recommendation systems may exhibit biases toward popular products, certain brands, or specific customer demographics. Businesses deploying recommendation systems should be aware of these ethical considerations and implement safeguards to ensure that their systems promote fair and diverse product discovery.

The catalogue coverage metric used in this study serves as a partial measure of recommendation diversity, and the finding that personalized models achieve dramatically higher coverage than popularity-based recommendations is encouraging from an ethical perspective, as it suggests that personalization can help counteract the concentration of attention on a few popular products.

## Annexure I: Summary of Key Numbers

The following table provides a quick reference to the key numbers and statistics reported throughout this dissertation.

| Metric                      | Value  |
|-----------------------------|--------|
| Total customers             | 1,000  |
| Total products              | 455    |
| Total interactions          | 49,058 |
| Explicit ratings            | 22,174 |
| Matrix sparsity             | 94.9%  |
| Training set size           | 17,739 |
| Test set size               | 4,434  |
| SVD components (optimal)    | 50     |
| SVD explained variance      | 40.1%  |
| Hybrid optimal alpha (RMSE) | 1.00   |
| Popularity RMSE             | 1.6557 |
| CF RMSE                     | 1.5748 |
| CB RMSE                     | 1.7107 |

| Metric                           | Value    |
|----------------------------------|----------|
| Hybrid RMSE                      | 1.5748   |
| Popularity Coverage              | 2.2%     |
| CF Coverage                      | 89.0%    |
| CB Coverage                      | 46.8%    |
| Hybrid Coverage                  | 85.3%    |
| H1 Chi-square statistic          | 30006.04 |
| H1 p-value                       | < 0.001  |
| VIP+Freq customer share          | 35.7%    |
| VIP+Freq purchase share          | 80.5%    |
| Coverage improvement (CF vs Pop) | 40x      |

## Annexure J: Plagiarism and AI Similarity Declaration

In compliance with the academic integrity guidelines of Sharda University, the following declaration is made regarding this dissertation:

**1. Plagiarism Similarity:** The content of this dissertation is original work prepared by the researcher. All sources consulted during the preparation of this document have been properly cited and referenced in accordance with the APA 7th Edition referencing style. The plagiarism similarity index is expected to be below 10%, in compliance with the university's academic integrity requirements.

**2. AI Similarity:** While data science tools and programming assistants were used for data generation, model development, and document formatting, the research design, analysis, interpretation, and conclusions presented in this dissertation represent the original intellectual contribution of the researcher. The AI similarity index is expected to be below 20%, in compliance with the university's guidelines.

**3. Data Authenticity:** The dataset used in this study is synthetic, generated programmatically for the purpose of this research. No real customer data has been used. The data generation process is fully documented in Chapter 3 and can be reproduced using the provided code and random seed.

**4. Code Originality:** The Python code used for data generation, model development, evaluation, and visualization is original work, built upon standard libraries (NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn, SciPy, ReportLab). No proprietary or

copyrighted code has been used.

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